

# Constructive Adversarial Fragility of Deep Neural Networks, ResNets, and Vision Transformers

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## Abstract

We connect the geometric balance-based view of adversarial vulnerability from [Shmalo26] with the constructive single-gradient-step framework of Daniely and Mehalel [DM26]. Our main observation is that balance plus concentration already yields the small-output part of the adversarial argument, so the remaining work is to verify gradient size and stability in each architecture. This leads to an abstract constructive theorem, a geometric theorem based on balance, transversality, and curvature, and architecture-specific derivative bounds for fixed-depth ResNets and simplified Vision Transformers, including deterministic standard-temperature sphere results for the latter. We also give multiclass, Gaussian, universality, and transfer extensions. The theorem-aligned CIFAR-10/CIFAR-100 experiments on CNNs, ResNets, and ViTs show that the proof quantities remain predictive after training, with conservative radius estimates and persistent local vulnerability signatures even in stronger ViT runs.

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\*Email: yitzchak.shmalo@gmail.com. Code for this work can be found at <https://github.com/yspennstate/adversarial-attacks>.

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## 1 Introduction

Adversarial examples—inputs that are perceptually indistinguishable from correctly classified points yet are misclassified by a trained model—pose a fundamental challenge to the deployment of deep neural networks in safety-critical applications [Szegedy14, GSS15, CW17]. A rich empirical literature has documented the phenomenon across vision [GSS15], language [Jia17], and reinforcement learning [Huang17], and a parallel theoretical effort seeks to explain *why* neural networks are generically vulnerable.

Two recent theoretical approaches have shed significant light on the problem.

**Geometric approach (our prior work).** In [Shmalo26], we proved that adversarial vulnerability is an *inevitable consequence of high-dimensional geometry* for any 1-Lipschitz,  $p$ -balanced binary classifier. The key tool is Lévy’s concentration of measure on the sphere: any Lipschitz function on  $\mathbb{S}^{d-1}(R)$  is  $O(R/\sqrt{d})$ -close to its median on a  $1 - \delta$  fraction of inputs. Since a balanced classifier has median near zero, its decision boundary is  $O(R/\sqrt{d})$ -close to most inputs. This result is *non-constructive*—it guarantees the existence of a nearby adversarial example but does not provide an algorithm to find one.

**Dynamical approach (Daniely–Mehalel).** Daniely and Mehalel [DM26] proved that for random convolutional neural networks with  $C^2$  activations, a *single gradient step* suffices to cross the decision boundary. Their proof is pointwise in a fixed input  $x$  with  $\|x\|_\infty \leq 1$ . It has two components: (i) a fixed-input small-output estimate  $|f(x)| = O(1)$  (their Lemma 8.1, proved via a direct moment calculation for their specific architecture), and (ii) a gradient lower/stability argument (their Sections 5–7, requiring detailed analysis of the CNN Jacobian).

**This paper: bridging the two.** We observe that an analogue of condition (i)—small output on a large set of inputs—is *not* architecture-specific once balance is known. It follows from the same concentration-of-measure argument that underlies our existence proof in [Shmalo26], and holds for *any* balanced, Lipschitz classifier. In many bias-free architectures with odd symmetry, balance itself is automatic under symmetric input laws. When that input law also satisfies a concentration theorem (for example, the uniform measure on the sphere, or

Gaussian measure), this feeds directly into a small-output estimate. Condition (ii), by contrast, is genuinely architecture-dependent and *cannot* be derived from balance alone (we give an explicit counterexample).

This separation clarifies the structure of the problem. We extract the constructive gradient-step argument into an abstract theorem (Theorem 4.1) with three conditions: (A1) small output, (A2) gradient lower bound, (A3) gradient stability. Condition (A1) is a consequence of the general geometric principle (Theorem 2.1). Conditions (A2) and (A3) must be verified architecture by architecture. At a geometric level, this leads to a model-independent constructive theorem (Theorem 4.9) showing that balance, transversality, and curvature are the only ingredients needed.

We then examine two architectures not covered in [DM26]:

- **Residual networks (ResNets):** Exact balance under symmetric input laws follows from oddness once the zero-logit set has measure zero, as in the symmetry analysis of [Shmalo26]. We prove that the  $1/\sqrt{t}$  residual scaling yields a deterministic full-Jacobian lower bound and a Hessian bound in the fixed-depth regime, which closes the one-step constructive argument once the geometric small-output estimate is inserted (Section 5.2).
- **Vision Transformers (ViTs):** We retain the simplified bias-free ViT balance mechanism of [Shmalo26, Section 5.4], and additionally prove a constructive theorem and sphere-based corollary in a high-temperature perturbative regime where the attention map is a controlled deformation of uniform attention. We also prove a deterministic standard-temperature compact-set theorem and a sphere-based corollary for fixed realized weights. The remaining open problem is to verify those standard-temperature constants automatically at random initialization (Section 5.3).

We further extend the results to multiclass classifiers (Section 6), Gaussian inputs (Section 7), and prove a universality theorem (Section 8).

The numerical results address the same local quantities that appear in the theory. Our main theorems are local and constructive: they depend on margin size, margin-gradient norm, curvature compatibility, and transfer slack rather than only on final classification accuracy. Accordingly, the relevant empirical test is not just robust accuracy or attack success, but the behavior of these proof quantities on trained CNN, ResNet, and ViT models. The CIFAR-10/CIFAR-100 experiments provide evidence that the constructive program developed in the theory sections continues to organize the behavior of trained networks rather than only asymptotic or initialization-level regimes.

**What is new versus [Shmalo26] and [DM26].** To be precise about the contributions:

1. Theorem 2.1 (balance implies small output) is new in its formulation, though it follows directly from the tools developed in [Shmalo26]. Its significance is that it replaces architecture-specific output estimates whenever balance can be verified independently.
2. The abstract constructive framework (Theorem 4.1) isolates the proof structure of [DM26] Section 4/Theorem 8.2. We do not claim novelty in the proof technique; rather, the contribution is the clean separation of geometric and architectural conditions.
3. Theorem 4.9, Proposition 5.1, and Corollary 5.2 are new geometric/model-theoretic statements: they separate the architecture-independent transversality/curvature principle from the architecture-specific task of verifying derivative bounds, and then reassemble them into a single constructive theorem for balanced smooth compositions.
4. Proposition 4.10 records, in the form needed here, a symmetry principle already present in [Shmalo26]. The genuinely new architecture-specific results in this paper are the fixed-depth ResNet constructive theorem (Proposition 5.6 and Corollary 5.7), the perturbative high-temperature ViT constructive results (Theorem 5.12 and Corollary 5.13), and the deterministic standard-temperature compact-set / sphere results for simplified ViTs together with their MLP-block extension (Theorem 5.14, Corollaries 5.15 and 5.16). The

ViT balance result (Proposition 5.10) itself is imported from [Shmalo26]; what remains open is the random-initialization verification problem of Remark 5.17.

5. The multiclass extension (Theorem 6.1), the accuracy-to-vulnerability theorem under density control (Theorem 3.1), the Gaussian extension (Theorem 7.1), the universality result (Theorem 8.1), and the transfer theorem (Theorem 8.3) do not appear in this form in [Shmalo26] or [DM26]; they are natural but useful extensions of the same concentration and local-geometry machinery.

## 1.1 Notation

We write  $\mathbb{S}^{d-1}(R)$  for the sphere of radius  $R$  in  $\mathbb{R}^d$ . When  $R = 1$  we simply write  $\mathbb{S}^{d-1}$ . The uniform probability measure on  $\mathbb{S}^{d-1}(R)$  is denoted  $\mu$ . For a function  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$ , a *median*  $m_f$  is any value satisfying  $\mu(f \geq m_f) \geq 1/2$  and  $\mu(f \leq m_f) \geq 1/2$ . When the median is not unique, we fix a choice; in the  $p = 1/2$  case below, we choose  $m_f = 0$ . We write  $\|\cdot\|$  for the  $\ell_2$  norm throughout unless otherwise specified. A binary classifier is a function  $\text{sign}(f)$  for some  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$ . We say  $\text{sign}(f)$  is *p-balanced* if  $\min(\mu(f > 0), \mu(f < 0)) \geq p$ , with  $p \leq 1/2$ . We write  $C_{\text{Lev}} = \sqrt{\pi/8}$  for the constant in Lévy's concentration inequality. We define

$$r(p) = \frac{R}{\sqrt{d-1}} \sqrt{2 \log(1/p)}, \quad \varepsilon_{\text{adv}}(p, \delta, d, R) = r(p) + \frac{R}{\sqrt{d-1}} \sqrt{2 \log(2C_{\text{Lev}}/\delta)}.$$

The quantity  $\varepsilon_{\text{adv}}$  is the adversarial radius from [Shmalo26]: for any 1-Lipschitz,  $p$ -balanced classifier, at least a  $1 - \delta$  fraction of inputs have an adversarial example within distance  $\varepsilon_{\text{adv}}$ .

## 2 Balance implies small output

The following theorem shows that the *same* concentration scale  $\varepsilon_{\text{adv}}$  that governs the adversarial radius also controls the magnitude of the classifier's output at typical inputs. This is the key geometric principle underlying the constructive framework.

**Theorem 2.1** (Balance implies small output). *Let  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  be 1-Lipschitz with respect to the extrinsic Euclidean metric (i.e.,  $|f(x) - f(y)| \leq \|x - y\|$  for all  $x, y \in \mathbb{S}^{d-1}(R)$ ); this implies 1-Lipschitz continuity with respect to the geodesic metric because  $\|x - y\| \leq d_{\text{geo}}(x, y)$  on the sphere). Suppose  $\text{sign}(f)$  is  $p$ -balanced for some  $p \in (0, 1/2]$ . Then for every  $\delta \in (0, 1)$ , a fraction at least  $1 - \delta$  of inputs  $x \in \mathbb{S}^{d-1}(R)$  (under the uniform measure  $\mu$ ) satisfy*

$$|f(x)| \leq \varepsilon_{\text{adv}}(p, \delta, d, R) = r(p) + \frac{R}{\sqrt{d-1}} \sqrt{2 \log\left(\frac{2C_{\text{Lev}}}{\delta}\right)}. \quad (1)$$

*Proof.* The proof has two steps: bounding the median, and applying concentration.

**Step 1: The median is small.** Since  $\text{sign}(f)$  is  $p$ -balanced, both  $\mu(f > 0) \geq p$  and  $\mu(f < 0) \geq p$ . We claim  $|m_f| \leq r(p)$ .

*Case  $p = 1/2$ .* By the definition of  $p$ -balance,  $\mu(f > 0) \geq 1/2$  and  $\mu(f < 0) \geq 1/2$ . Then  $\mu(f \geq 0) \geq 1/2$  and  $\mu(f \leq 0) \geq 1/2$ , so  $m_f = 0$  is a valid median. We fix this choice; then  $|m_f| = 0 \leq r(1/2)$ .

*Case  $p < 1/2$ .* Suppose  $m_f > 0$  (the case  $m_f < 0$  is symmetric). Then  $\mu(f < 0) \leq \mu(f - m_f < -m_f) \leq C_{\text{Lev}} \exp(-(d-1)m_f^2/(2R^2))$ , where we use the one-sided form of Lévy's inequality from Lemma 2.3. Since  $\mu(f < 0) \geq p$ , we get  $C_{\text{Lev}} \exp(-(d-1)m_f^2/(2R^2)) \geq p$ , so

$$m_f \leq \frac{R}{\sqrt{d-1}} \sqrt{2 \log(C_{\text{Lev}}/p)} \leq \frac{R}{\sqrt{d-1}} \sqrt{2 \log(1/p)} = r(p),$$

where the last inequality uses  $C_{\text{Lev}} < 1$ . Thus  $|m_f| \leq r(p)$  in all cases.

**Step 2: Concentration around the median.** By Lévy’s inequality (Lemma 2.3),

$$\mu(|f - m_f| > t) \leq 2C_{\text{Lev}} \exp\left(-\frac{(d-1)t^2}{2R^2}\right).$$

Setting the right side equal to  $\delta$  and solving for  $t$ :

$$t = \frac{R}{\sqrt{d-1}} \sqrt{2\log(2C_{\text{Lev}}/\delta)}.$$

On the event  $|f(x) - m_f| \leq t$  (which has measure  $\geq 1 - \delta$ ), we have

$$|f(x)| \leq |m_f| + |f(x) - m_f| \leq r(p) + t = \varepsilon_{\text{adv}}(p, \delta, d, R). \quad \square$$

**Corollary 2.2** (Small output for  $L$ -Lipschitz classifiers). *Let  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  be  $L$ -Lipschitz with respect to the extrinsic Euclidean metric, and suppose  $\text{sign}(f)$  is  $p$ -balanced for some  $p \in (0, 1/2]$ . Then for every  $\delta \in (0, 1)$ , a fraction at least  $1 - \delta$  of inputs satisfy*

$$|f(x)| \leq L \varepsilon_{\text{adv}}(p, \delta, d, R).$$

*Proof.* If  $L = 0$ , then  $f$  is constant and the conclusion is trivial. Otherwise apply Theorem 2.1 to  $\tilde{f} = f/L$ , which is 1-Lipschitz and has the same sign pattern as  $f$ .  $\square$

**Lemma 2.3** (Lévy’s concentration inequality on the sphere; see [Ledoux01, Chapter 2]). *Let  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  be 1-Lipschitz and let  $m_f$  denote its median. Then for all  $t > 0$ ,*

$$\mu(f - m_f > t) \leq C_{\text{Lev}} \exp\left(-\frac{(d-1)t^2}{2R^2}\right), \quad \mu(m_f - f > t) \leq C_{\text{Lev}} \exp\left(-\frac{(d-1)t^2}{2R^2}\right),$$

and therefore

$$\mu(|f - m_f| > t) \leq 2C_{\text{Lev}} \exp\left(-\frac{(d-1)t^2}{2R^2}\right),$$

where  $C_{\text{Lev}} = \sqrt{\pi/8}$ .

**Remark 2.4** (The key insight). Theorem 2.1 and the adversarial existence theorem of [Shmalo26] are controlled by the *same quantity*  $\varepsilon_{\text{adv}}$ . For a  $1 - \delta$  fraction of inputs, the adversarial radius is at most  $\varepsilon_{\text{adv}}$  (there exists a class-changing perturbation within distance  $\varepsilon_{\text{adv}}$ ), and the output magnitude is at most  $\varepsilon_{\text{adv}}$  (the function value is small). This is not a coincidence: both are consequences of the single fact that concentration forces a Lipschitz function to be  $O(R/\sqrt{d})$ -close to its median, and balance forces the median to be near zero.

This observation replaces architecture-specific small-output estimates once balance has been established by other means (for example, by odd symmetry under a symmetric input law together with a concentrated ambient measure; see Proposition 4.10 below). Theorem 2.1 applies to *any* 1-Lipschitz balanced classifier, regardless of architecture.

### 3 Adversarial fragility of accurate classifiers

The results so far apply to classifiers under the uniform measure  $\mu$  on the sphere. In practice, classifiers are trained on a data distribution  $\nu$  that is far from uniform. We now prove that *any classifier with nontrivial accuracy on balanced data is adversarially fragile*, with an explicit adversarial radius depending on the accuracy, label balance, density ratio, number of classes, and dimension. This result applies to arbitrary measurable classifiers without assuming Lipschitz logits, random initialization, or specific architecture.

**Theorem 3.1** (Adversarial fragility of accurate classifiers). *Let  $F : \mathbb{S}^{d-1}(R) \rightarrow [K]$  be a measurable  $K$ -class classifier. Let  $\nu$  be a probability measure on  $\mathbb{S}^{d-1}(R)$  that is absolutely continuous with respect to  $\mu$ , with density bounded by  $\rho_{\max}$ :  $d\nu/d\mu \leq \rho_{\max}$ . Suppose  $F$  achieves accuracy  $\alpha$  under  $\nu$ , meaning  $\nu(F(x) = y(x)) \geq \alpha$  where  $y : \mathbb{S}^{d-1}(R) \rightarrow [K]$  is the ground-truth labeling.*

*Assume the ground truth  $y$  is  $q$ -balanced under  $\nu$ :  $\nu(y = k) \geq q$  for all  $k$ , with  $q \in (0, 1/K]$ . Assume  $\alpha > 1 - q$ . Then:*

- (i)  *$F$  is  $p$ -balanced under  $\mu$  with  $p \geq (\alpha + q - 1)/\rho_{\max}$ .*
- (ii) *For every  $\delta > 0$ , a fraction at least  $1 - \rho_{\max}\delta$  of inputs under  $\nu$  satisfy: there exists  $y$  with  $F(y) \neq F(x)$  and*

$$\|y - x\| \leq r(p) + \frac{R}{\sqrt{d-1}} \sqrt{2 \log \left( \frac{KC_{\text{Lev}}}{\delta} \right)}, \quad (2)$$

where  $p = (\alpha + q - 1)/\rho_{\max}$ .

**Proof. Part (i): Accuracy implies balance.** For any class  $k$ :

$$\nu(F = k) \geq \nu(F = k, y = k) \geq \nu(y = k) - \nu(F \neq y) \geq q - (1 - \alpha) = \alpha + q - 1.$$

The first inequality holds since  $\{F = k, y = k\} \subseteq \{F = k\}$ ; the second uses  $\nu(F = k, y = k) = \nu(y = k) - \nu(F \neq k, y = k) \geq \nu(y = k) - \nu(F \neq y)$ . The assumption  $\alpha > 1 - q$  ensures  $\alpha + q - 1 > 0$ . For the  $\mu$ -balance: since  $\nu \leq \rho_{\max} \mu$ , we have  $\mu(F = k) \geq \nu(F = k)/\rho_{\max} \geq (\alpha + q - 1)/\rho_{\max} = p$ .

**Part (ii): Adversarial radius via set thickening.** By part (i), each predicted-class region  $A_k = \{x : F(x) = k\}$  has  $\mu(A_k) \geq p$ . A point  $x$  with  $F(x) = j$  has no adversarial neighbor within radius  $\varepsilon$  only if  $x \notin (A_k)_{\varepsilon}$  for every  $k \neq j$ , where  $(A_k)_{\varepsilon}$  denotes the  $\varepsilon$ -thickening of  $A_k$ . Therefore

$$\mu(\text{no adversarial neighbor within } \varepsilon) \leq \sum_{k=1}^K \mu(A_{k,\varepsilon}^c).$$

By Lévy’s isoperimetric inequality on the sphere [Ledoux01], for any measurable  $A \subseteq \mathbb{S}^{d-1}(R)$  with  $\mu(A) \geq p$ ,

$$\mu(A_{\varepsilon}^c) \leq C_{\text{Lev}} \exp \left( - \frac{(d-1)(\varepsilon - r(p))_+^2}{2R^2} \right).$$

Setting  $\varepsilon = r(p) + \frac{R}{\sqrt{d-1}} \sqrt{2 \log(KC_{\text{Lev}}/\delta)}$  makes each term  $\leq \delta/K$ , so the sum is  $\leq \delta$ . Since  $\nu \leq \rho_{\max} \mu$ , the event “no adversarial neighbor within  $\varepsilon$ ” has  $\nu$ -measure  $\leq \rho_{\max} \delta$ .  $\square$

**Remark 3.2** (Interpretation). Theorem 3.1 makes the folklore intuition “accurate classifiers are adversarially fragile” precise and quantitative. Note that this result does not require Lipschitz logits—it applies to any measurable classifier. For a  $K = 10$  class problem in dimension  $d = 3072$  (CIFAR-10 images) with inputs of norm  $R \approx \sqrt{d} \approx 55$ , accuracy  $\alpha = 0.95$ , and balanced ground truth  $q = 1/K = 0.1$ :  $p = 0.05/\rho_{\max}$ , and the adversarial radius is approximately  $\frac{55}{\sqrt{3071}} (\sqrt{2 \log(20)} + \sqrt{2 \log(10)}) \approx 4.2$  for  $\rho_{\max} = 1$ . The bound requires  $\alpha > 1 - q = 0.9$  for  $K = 10$ ; for  $\alpha = 0.9$  the balance parameter  $p$  degenerates to 0.

## 4 The constructive gradient-step framework

Knowing that  $|f(x_0)|$  is small at a typical input  $x_0$  is not, by itself, sufficient to *construct* an adversarial example. One needs additional information about the local geometry of  $f$  near  $x_0$ —specifically, that the gradient is large and stable enough to guarantee that a step in the gradient direction crosses the decision boundary.

The following theorem abstracts the constructive argument from [DM26]. The proof follows their Theorem 8.2 / Section 4 argument; we reproduce it here for completeness and to make explicit which conditions are used where.

**Theorem 4.1** (Constructive adversarial perturbation). *Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be differentiable, and let  $x_0 \in \mathbb{R}^d$  with  $f(x_0) \neq 0$  satisfy the following three conditions:*

(A1) **Small output:**  $|f(x_0)| \leq a$  for some  $a > 0$ .

(A2) **Gradient lower bound:**  $\|\nabla f(x_0)\| \geq C_g$  for some  $C_g > 0$ .

(A3) **Gradient stability:** For all  $x'$  with  $\|x' - x_0\| \leq 2a/C_g$ :  $\|\nabla f(x') - \nabla f(x_0)\| \leq C_g/3$ .

Define

$$x_1 = x_0 - \text{sign}(f(x_0)) \frac{2a}{\|\nabla f(x_0)\|^2} \nabla f(x_0).$$

Then  $\text{sign}(f(x_1)) \neq \text{sign}(f(x_0))$  and  $\|x_1 - x_0\| \leq 2a/C_g$ .

*Proof.* Set  $\sigma_0 := \text{sign}(f(x_0)) \in \{\pm 1\}$  and  $g := \nabla f(x_0)$ . Define the step size  $s^* = 2a/\|g\|^2$ , so that  $x_1 = x_0 - \sigma_0 s^* g$ .

**Step size bound.**  $\|x_1 - x_0\| = s^* \|g\| = 2a/\|g\| \leq 2a/C_g$ , using (A2).

**Sign change.** Consider the signed path

$$\varphi(s) := \sigma_0 f(x_0 - \sigma_0 s g), \quad s \in [0, s^*].$$

By the fundamental theorem of calculus,

$$\varphi(s^*) = \varphi(0) - \int_0^{s^*} \langle \nabla f(x_0 - \sigma_0 s g), g \rangle ds.$$

For any  $s \in [0, s^*]$ , the point  $x_0 - \sigma_0 s g$  satisfies

$$\|(x_0 - \sigma_0 s g) - x_0\| = s \|g\| \leq s^* \|g\| = \frac{2a}{\|g\|} \leq \frac{2a}{C_g}.$$

Hence by (A3),  $\|\nabla f(x_0 - \sigma_0 s g) - g\| \leq C_g/3$ . Therefore

$$\langle \nabla f(x_0 - \sigma_0 s g), g \rangle = \|g\|^2 + \langle \nabla f(x_0 - \sigma_0 s g) - g, g \rangle \geq \|g\|^2 - \frac{C_g}{3} \|g\| \geq \|g\|^2 - \frac{\|g\|}{3} \|g\| = \frac{2}{3} \|g\|^2,$$

where in the second inequality we used  $C_g \leq \|g\|$  (which follows from (A2)). Thus

$$\varphi(s^*) \leq \varphi(0) - \frac{2}{3} \|g\|^2 s^* \leq |f(x_0)| - \frac{2}{3} \|g\|^2 \cdot \frac{2a}{\|g\|^2} \leq a - \frac{4a}{3} = -\frac{a}{3} < 0.$$

Since  $\varphi(s^*) = \sigma_0 f(x_1) < 0$ , we have  $\text{sign}(f(x_1)) = -\sigma_0 = -\text{sign}(f(x_0))$ .  $\square$

*Remark 4.2* (Positive rescaling leaves the attack unchanged). If  $\tilde{f} = cf$  for some  $c > 0$ , then  $\text{sign}(\tilde{f}) = \text{sign}(f)$  and the one-step point from Theorem 4.1 is unchanged:

$$x_0 - \text{sign}(cf(x_0)) \frac{2ca}{\|\nabla(cf)(x_0)\|^2} \nabla(cf)(x_0) = x_0 - \text{sign}(f(x_0)) \frac{2a}{\|\nabla f(x_0)\|^2} \nabla f(x_0).$$

Moreover the quantities  $(a, C_g, H)$  scale to  $(ca, cC_g, cH)$ , so the compatibility inequality  $a \leq C_g^2/(6H)$  is invariant under positive rescaling. This is why one may freely normalize logits when applying Theorem 2.1.

**Lemma 4.3** (Hessian control implies gradient stability). *Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be  $C^2$ , let  $x_0 \in \mathbb{R}^d$ , and let  $r > 0$ . Assume that*

$$\sup_{\|z - x_0\| \leq r} \|\nabla^2 f(z)\|_{\text{op}} \leq H.$$

Then for every  $x'$  with  $\|x' - x_0\| \leq r$ ,

$$\|\nabla f(x') - \nabla f(x_0)\| \leq H \|x' - x_0\|.$$

In particular, condition (A3) holds whenever  $r = 2a/C_g$  and  $H \leq C_g^2/(6a)$ .

*Proof.* Write  $\Delta = x' - x_0$ . By the fundamental theorem of calculus,

$$\nabla f(x') - \nabla f(x_0) = \int_0^1 \nabla^2 f(x_0 + s\Delta) \Delta ds.$$

Hence

$$\|\nabla f(x') - \nabla f(x_0)\| \leq \int_0^1 \|\nabla^2 f(x_0 + s\Delta)\|_{\text{op}} \|\Delta\| ds \leq H \|\Delta\|.$$

If  $r = 2a/C_g$  and  $H \leq C_g^2/(6a)$ , then for every  $\|x' - x_0\| \leq 2a/C_g$ :

$$\|\nabla f(x') - \nabla f(x_0)\| \leq H \cdot \frac{2a}{C_g} \leq \frac{C_g}{3},$$

which is exactly (A3).  $\square$

*Remark 4.4* (Program for a one-step adversarial proof). Theorem 4.1 and Lemma 4.3 reduce a constructive adversarial theorem for a given architecture to four tasks:

- (P1) prove a small-output bound  $|f(x_0)| \leq a$  on a large set of inputs, typically from balance plus Theorem 2.1;
- (P2) prove a gradient lower bound  $\|\nabla f(x_0)\| \geq C_g$ ;
- (P3) prove a local Hessian bound  $\sup_{\|x-x_0\| \leq 2a/C_g} \|\nabla^2 f(x)\|_{\text{op}} \leq H$ ;
- (P4) verify the compatibility inequality  $a \leq C_g^2/(6H)$ .

Once these are in place, the explicit one-step attack is

$$x_1 = x_0 - \text{sign}(f(x_0)) \frac{2a}{\|\nabla f(x_0)\|^2} \nabla f(x_0),$$

and Theorem 4.1 guarantees  $\text{sign}(f(x_1)) \neq \text{sign}(f(x_0))$ . Sections 5.2 and 5.3 follow exactly this program.

*Remark 4.5* (Which conditions are geometric and which are architectural). Condition (A1) is a consequence of Theorem 2.1: for any 1-Lipschitz,  $p$ -balanced classifier, we may take  $a = \varepsilon_{\text{adv}}(p, \delta, d, R)$  and the condition holds for a  $1 - \delta$  fraction of inputs. This is purely geometric—it depends only on balance, Lipschitz continuity, and the dimension.

Conditions (A2) and (A3), by contrast, are *genuinely architecture-dependent* and *cannot* be deduced from balance alone.

*Example 4.6* (Balance does not imply gradient lower bound). Consider  $f : [-1, 1] \rightarrow \mathbb{R}$  defined by  $f(t) = t^3$ . Under the uniform measure on  $[-1, 1]$ , this function is exactly balanced:  $\mu(f > 0) = \mu(f < 0) = 1/2$ . However,  $f'(0) = 0$ , so the gradient lower bound fails at  $t_0 = 0$ .

More generally, a balanced smooth function may be locally flat at some inputs where its value is small. This shows that verifying (A2) requires information about the specific architecture, not just the classifier’s balance properties.

*Remark 4.7* (Connection to [DM26]). In [DM26], condition (A1) is their Lemma 8.1, proved by a moment calculation specific to their CNN architecture. Conditions (A2) and (A3) are verified in their Sections 5–7 through singular-value control for random convolutional operators, layerwise Jacobian/Frobenius lower bounds, and a local gradient-robustness estimate. Our contribution here is to show that (A1) holds much more generally once balance is known, and to isolate the precise gradient-analysis tasks that remain for additional architectures.

*Remark 4.8* (Sphere versus ambient space). Theorem 4.1 is stated for  $f : \mathbb{R}^d \rightarrow \mathbb{R}$ , and the step  $x_1 = x_0 - \eta \text{sign}(f(x_0)) \nabla f(x_0)$  generally leaves the sphere  $\mathbb{S}^{d-1}(R)$ . When the logit function  $f$  is defined on the sphere, we interpret Theorem 4.1 as applying to any  $C^1$  extension  $\tilde{f} : \mathbb{R}^d \rightarrow \mathbb{R}$  of  $f$  to a neighborhood of  $\mathbb{S}^{d-1}(R)$ , with the understanding that the adversarial perturbation  $x_1$  lies in ambient  $\mathbb{R}^d$  rather than on the sphere. For neural networks, which are naturally defined on



$\mathbb{R}^d$ , this is immediate: the restriction to  $\mathbb{S}^{d-1}(R)$  is for the concentration-of-measure argument (Theorem 2.1), while the gradient step uses the ambient Euclidean structure. An alternative intrinsic formulation using the Riemannian gradient and the exponential map on  $\mathbb{S}^{d-1}(R)$  is possible but notationally heavier; we do not pursue it here.

**Theorem 4.9** (Constructive vulnerability from transversality and curvature). *Let  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  be an  $L$ -Lipschitz logit whose sign classifier is  $p$ -balanced. Let  $\tilde{f}$  be a  $C^2$  extension of  $f$  to the Euclidean neighborhood*

$$U_\rho := \{z \in \mathbb{R}^d : \text{dist}(z, \mathbb{S}^{d-1}(R)) \leq \rho\}.$$

*Assume there are constants  $m, H > 0$  such that*

$$\|\nabla \tilde{f}(x)\| \geq m \quad \text{for every } x \in \mathbb{S}^{d-1}(R), \quad \sup_{z \in U_\rho} \|\nabla^2 \tilde{f}(z)\|_{\text{op}} \leq H,$$

*where*

$$\rho := \frac{2L \varepsilon_{\text{adv}}(p, \delta, d, R)}{m}.$$

*Assume moreover that  $\mu(f = 0) = 0$  on  $\mathbb{S}^{d-1}(R)$ . If*

$$L \varepsilon_{\text{adv}}(p, \delta, d, R) \leq \frac{m^2}{6H},$$

*then for a fraction at least  $1 - \delta$  of inputs  $x_0 \in \mathbb{S}^{d-1}(R)$ , the single gradient step*

$$x_1 = x_0 - \text{sign}(f(x_0)) \frac{2L \varepsilon_{\text{adv}}(p, \delta, d, R)}{\|\nabla \tilde{f}(x_0)\|^2} \nabla \tilde{f}(x_0)$$

*satisfies  $\text{sign}(\tilde{f}(x_1)) \neq \text{sign}(f(x_0))$  and*

$$\|x_1 - x_0\| \leq \frac{2L \varepsilon_{\text{adv}}(p, \delta, d, R)}{m}.$$

*Proof.* By Corollary 2.2, a  $1 - \delta$  fraction of inputs  $x_0 \in \mathbb{S}^{d-1}(R)$  satisfy

$$|f(x_0)| \leq L \varepsilon_{\text{adv}}(p, \delta, d, R).$$

By the additional hypothesis  $\mu(f = 0) = 0$ , we may discard a null set and assume  $f(x_0) \neq 0$ . For such an  $x_0$ , let  $a = L \varepsilon_{\text{adv}}(p, \delta, d, R)$  and  $C_g = m$ . The gradient lower bound gives condition (A2), and the Hessian bound on  $U_\rho$  together with Lemma 4.3 gives condition (A3), because  $\rho = 2a/C_g$  and  $a \leq C_g^2/(6H)$  by assumption. Applying Theorem 4.1 to  $\tilde{f}$  proves the claim.  $\square$

**Proposition 4.10** (Odd bias-free architectures are exactly balanced; following [Shmalo26, Sections 4–5]). *Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be realized by a network built from bias-free linear maps (including convolutions), pointwise odd activations, residual additions of the form  $h \mapsto h + g(h)$ , and a final bias-free linear readout. Then  $f(-x) = -f(x)$  for all  $x \in \mathbb{R}^d$ . Consequently, under any input distribution  $\mu$  symmetric about the origin and satisfying  $\mu(f = 0) = 0$ , the classifier  $\text{sign}(f)$  is exactly  $\frac{1}{2}$ -balanced under  $\mu$ .*

*Proof.* Bias-free linear maps are odd, pointwise odd activations are odd, the composition of odd maps is odd, and if  $g$  is odd then so is  $h \mapsto h + g(h)$ . By induction through the network graph, every hidden representation is odd in the input, and hence the final scalar readout is odd as well. If  $X \stackrel{d}{=} -X$ , then  $f(X) \stackrel{d}{=} -f(X)$ , so the law of  $f(X)$  is symmetric about 0. If moreover  $\mu(f = 0) = 0$ , this symmetry implies  $\mu(f > 0) = \mu(f < 0) = 1/2$ .  $\square$

## 5 Verifying gradient conditions for deep architectures

In this section we separate what is rigorous from what remains conditional for three architecture families at random initialization: fully-connected/convolutional networks (via [DM26]), residual networks, and Vision Transformers. Condition (A1) is handled separately: for odd, bias-free architectures under the *uniform* measure on the sphere it follows from Proposition 4.10 and Theorem 2.1, while for the general smooth-CNN setting of [DM26] we retain their original small-output lemma.

**Proposition 5.1** (Composition bounds for smooth deep maps). *Let  $\Phi_\ell : \mathbb{R}^m \rightarrow \mathbb{R}^m$  be  $C^2$  maps for  $\ell = 1, \dots, t$ , and define*

$$h_0(x) = x, \quad h_\ell(x) = \Phi_\ell(h_{\ell-1}(x)), \quad F(x) = h_t(x).$$

*Assume that for every  $\ell$  there are constants  $\alpha_\ell, \nu_\ell, H_\ell \geq 0$  such that for all  $u \in \mathbb{R}^m$ ,*

$$\|D\Phi_\ell(u)\|_{\text{op}} \leq \alpha_\ell, \quad \text{smin}(D\Phi_\ell(u)) \geq \nu_\ell, \quad \|D^2\Phi_\ell(u)\|_{\text{op}} \leq H_\ell.$$

*Then for every  $x \in \mathbb{R}^m$ :*

(i)

$$\|DF(x)\|_{\text{op}} \leq \prod_{\ell=1}^t \alpha_\ell, \quad \text{smin}(DF(x)) \geq \prod_{\ell=1}^t \nu_\ell.$$

(ii)

$$\|D^2F(x)\|_{\text{op}} \leq \sum_{\ell=1}^t H_\ell \left( \prod_{j<\ell} \alpha_j \right)^2 \left( \prod_{j>\ell} \alpha_j \right).$$

*Consequently, for any readout vector  $w \in \mathbb{R}^m$ , the scalar logit  $f(x) = \langle w, F(x) \rangle$  satisfies*

$$\|\nabla f(x)\| \geq \|w\| \prod_{\ell=1}^t \nu_\ell, \quad \|\nabla^2 f(x)\|_{\text{op}} \leq \|w\| \sum_{\ell=1}^t H_\ell \left( \prod_{j<\ell} \alpha_j \right)^2 \left( \prod_{j>\ell} \alpha_j \right).$$

*If moreover  $\alpha_\ell \leq \alpha$  and  $H_\ell \leq H$  for every  $\ell$ , then*

$$\|\nabla^2 f(x)\|_{\text{op}} \leq \|w\| tH \alpha^{2t-2}.$$

*Proof.* Set  $J_\ell(x) = Dh_\ell(x)$  and  $B_\ell(x) = D^2h_\ell(x)$ . By the chain rule,

$$J_\ell(x) = D\Phi_\ell(h_{\ell-1}(x)) J_{\ell-1}(x).$$

Thus

$$\|J_\ell(x)\|_{\text{op}} \leq \alpha_\ell \|J_{\ell-1}(x)\|_{\text{op}}, \quad \text{smin}(J_\ell(x)) \geq \nu_\ell \text{smin}(J_{\ell-1}(x)).$$

Starting from  $J_0 = I$ , iteration gives part (i).

For the second derivative, the chain rule gives, for every  $u, v \in \mathbb{R}^m$ ,

$$B_\ell(x)[u, v] = D^2\Phi_\ell(h_{\ell-1}(x))[J_{\ell-1}(x)u, J_{\ell-1}(x)v] + D\Phi_\ell(h_{\ell-1}(x)) B_{\ell-1}(x)[u, v].$$

Taking operator norms,

$$\|B_\ell(x)\|_{\text{op}} \leq H_\ell \|J_{\ell-1}(x)\|_{\text{op}}^2 + \alpha_\ell \|B_{\ell-1}(x)\|_{\text{op}}.$$

Since part (i) gives  $\|J_{\ell-1}(x)\|_{\text{op}} \leq \prod_{j<\ell} \alpha_j$ , we obtain the recursion

$$\|B_\ell(x)\|_{\text{op}} \leq H_\ell \left( \prod_{j<\ell} \alpha_j \right)^2 + \alpha_\ell \|B_{\ell-1}(x)\|_{\text{op}}.$$

Starting from  $B_0 = 0$  and iterating this recursion yields

$$\|D^2F(x)\|_{\text{op}} = \|B_t(x)\|_{\text{op}} \leq \sum_{\ell=1}^t H_\ell \left( \prod_{j<\ell} \alpha_j \right)^2 \left( \prod_{j>\ell} \alpha_j \right).$$

Finally, if  $f(x) = \langle w, F(x) \rangle$ , then  $\nabla f(x) = DF(x)^T w$  and  $\nabla^2 f(x) = w^T D^2F(x)$ , so the stated scalar bounds follow immediately. The simplified bound under uniform  $\alpha, H$  is obtained by observing that each summand is at most  $H\alpha^{2t-2}$ .  $\square$

**Corollary 5.2** (Constructive theorem for balanced smooth compositions). *Let  $F$  and  $f(x) = \langle w, F(x) \rangle$  be as in Proposition 5.1. Assume inputs are drawn from the uniform measure on  $\mathbb{S}^{m-1}(R) \subset \mathbb{R}^m$ , that  $\text{sign}(f)$  is  $p$ -balanced there, and that  $\mu(f=0) = 0$ . Set*

$$L_F := \|w\| \prod_{\ell=1}^t \alpha_\ell, \quad m_F := \|w\| \prod_{\ell=1}^t \nu_\ell,$$

and

$$H_F := \|w\| \sum_{\ell=1}^t H_\ell \left( \prod_{j<\ell} \alpha_j \right)^2 \left( \prod_{j>\ell} \alpha_j \right).$$

If

$$6H_F L_F \varepsilon_{\text{adv}}(p, \delta, m, R) \leq m_F^2,$$

then for a  $1 - \delta$  fraction of inputs  $x_0 \in \mathbb{S}^{m-1}(R)$ , the single gradient step

$$x_1 = x_0 - \text{sign}(f(x_0)) \frac{2L_F \varepsilon_{\text{adv}}(p, \delta, m, R)}{\|\nabla f(x_0)\|^2} \nabla f(x_0)$$

satisfies  $\text{sign}(f(x_1)) \neq \text{sign}(f(x_0))$  and

$$\|x_1 - x_0\| \leq \frac{2L_F \varepsilon_{\text{adv}}(p, \delta, m, R)}{m_F}.$$

*Proof.* Proposition 5.1 gives the global bounds

$$\|\nabla f(x)\| \geq m_F, \quad \|\nabla^2 f(x)\|_{\text{op}} \leq H_F$$

for every  $x \in \mathbb{R}^m$ , and also  $\|Df(x)\|_{\text{op}} \leq L_F$ . Hence  $f$  is  $L_F$ -Lipschitz on the sphere. The displayed compatibility inequality is exactly  $L_F \varepsilon_{\text{adv}} \leq m_F^2/(6H_F)$ . Therefore Theorem 4.9 applies with  $\tilde{f} = f$ ,  $L = L_F$ ,  $m = m_F$ , and  $H = H_F$ .  $\square$

## 5.1 Fully-connected and convolutional networks

For completeness, we recall the relevant prior results.

**Theorem 5.3** (Constructive attacks for random CNNs; faithful summary of [DM26, Theorem 3.1, proved in Section 8]). *Let  $F$  be a random convolutional neural network satisfying the assumptions of [DM26, Section 2]:  $C^2$  activation, constant depth, controlled widths, Xavier initialization, and the pointwise nondegeneracy condition*

$$\sigma'(r)^2 + \sigma'(-r)^2 \geq 2c_{\text{DM}}^2 \quad \text{for all } r \in \mathbb{R}.$$

Fix an input  $x \in L^2(G, \mathbb{R}^{d_0})$  with  $\|x\|_\infty \leq 1$ . Then, with high probability over the random weights:

1.  $|F(x)| = O(1)$ ; this is their fixed-input small-output estimate [DM26, Lemma 8.1].

2. If  $J_H(x)$  denotes the feature Jacobian of the convolutional part, then

$$\|J_H(x)\|_F^2 \geq \Omega((0.1c_{\text{DM}}^2)^t d_t |G|)$$

by [DM26, Theorem 6.4]. After the final Gaussian readout, this implies the scalar gradient lower bound

$$\|\nabla F(x)\| = \Omega(1)$$

by [DM26, Lemma 6.6].

3. There is a radius  $\rho = O(1)$  such that for all  $x' \in B(x, \rho)$ ,

$$\|\nabla F(x') - \nabla F(x)\| \leq \|\nabla F(x)\| / 3;$$

this is the local gradient-robustness step from [DM26, Lemma 7.4 and Section 8].

Consequently, [DM26, Theorem 8.2] yields a single gradient-descent step of Euclidean length  $O(1)$  that reaches some  $x_1$  with  $\text{sign}(F(x_1)) \neq \text{sign}(F(x))$ .

*Remark 5.4* (Fully-connected networks). Constructive adversarial vulnerability for random fully-connected ReLU networks was established earlier by Daniely and Shacham [DanielyS20]. For smooth fully-connected nets, [DM26] is best viewed as a source of proof ideas (Jacobian/Frobenius lower bounds plus a gradient-robustness step), not as a literal  $|G| = 1$  specialization of the uploaded CNN theorem. We state Theorem 5.3 for CNNs specifically because the convolutional setting is what [DM26] actually proves.

*Remark 5.5* (Stronger fully-connected results in the literature). For fully-connected random networks, the existing literature is already stronger than the minimal discussion needed here. Bartlett, Bubeck, and Cherapanamjeri [BartlettBC21] analyze multi-layer random ReLU networks, while Bubeck, Cherapanamjeri, Gidel, and Tachet des Combes [BubeckCGT21] prove that a single gradient step finds adversarial examples in overcomplete two-layer random networks and, more generally, in subexponential-width random networks with smooth activations. Because those results are architecture-specific and already cover the fully-connected case well, the present paper focuses its new constructive analysis on the ResNet and ViT settings instead.

The proof in [DM26] is pointwise in a fixed input and proceeds by analyzing the feature Jacobian layer by layer. The key ingredients are the singular-value analysis of random convolutional operators from their Section 5, the Rademacher identity  $\|J_H(x)\|_F^2 = \mathbb{E}_r[\|J_H(x)r\|^2]$  used in their Theorem 6.4, the passage from feature-Jacobian Frobenius norm to the scalar output gradient after the final Gaussian readout in their Lemma 6.6, and the matrix-concentration argument behind their Lemma 7.4. In the odd, bias-free subfamily under the *uniform* sphere measure, condition (A1) can alternatively be obtained from Proposition 4.10 and Theorem 2.1. We refer the reader to [DM26] for the full proof.

## 5.2 Residual networks

We consider a residual network with  $t$  blocks of the form

$$h_\ell = h_{\ell-1} + \frac{1}{\sqrt{t}} \sigma(\Lambda_\ell h_{\ell-1}), \quad \ell = 1, \dots, t, \quad (3)$$

where  $\Lambda_\ell \in \mathbb{R}^{d_0 \times d_0}$  are weight matrices with Xavier initialization ( $(\Lambda_\ell)_{ij} \sim \mathcal{N}(0, 1/d_0)$ ),  $\sigma$  is a  $C^2$  odd activation with  $\sigma(0) = 0$ ,  $\|\sigma'\|_\infty \leq 1$ ,  $c^2 := \mathbb{E}[\sigma'(Z)^2] > 0$  for  $Z \sim \mathcal{N}(0, 1)$ , and we assume constant width  $d_\ell = d_0$  for all  $\ell$ . The oddness assumption (i.e.,  $\sigma(-u) = -\sigma(u)$ , equivalently  $\sigma'$  is even) is used in the cross-term cancellation below; tanh and odd-shifted softplus are examples. The  $1/\sqrt{t}$  scaling of the residual branch is standard practice [HeZRS16] and ensures that each layer contributes an  $O(1/\sqrt{t})$  correction. The final output is  $f(x) = w^T h_t$  for a readout vector  $w \sim \mathcal{N}(0, N^{-1}I_N)$  with  $N = d_0$ . By Proposition 4.10, this ResNet is exactly balanced under any symmetric input law for which  $\mu(f = 0) = 0$ .

**Proposition 5.6** (Fixed-depth ResNet Jacobian/Hessian bounds). *Let  $f$  be a random ResNet as above (in particular,  $\sigma$  is odd and  $C^2$ ). Set*

$$M := \max_{1 \leq \ell \leq t} \|\Lambda_\ell\|_{\text{op}}, \quad S_1 := \|\sigma'\|_\infty, \quad S_2 := \|\sigma''\|_\infty,$$

and define

$$\alpha := 1 + \frac{S_1 M}{\sqrt{t}}, \quad \nu := 1 - \frac{S_1 M}{\sqrt{t}}.$$

Assume  $\nu > 0$ . Then the following hold. Parts (a)–(c) are deterministic conditional on the realized weights, while part (d) is an expectation over initialization.

(a) For every  $x \in \mathbb{R}^{d_0}$ , each block Jacobian

$$A_\ell(x) := I + \frac{1}{\sqrt{t}} D_\ell(x) \Lambda_\ell, \quad D_\ell(x) := \text{diag}(\sigma'(\Lambda_\ell h_{\ell-1}(x))),$$

satisfies

$$s_{\min}(A_\ell(x)) \geq \nu, \quad \|A_\ell(x)\|_{\text{op}} \leq \alpha.$$

(b) Let  $J_f(x) = \partial h_t(x)/\partial x$  denote the input-to-feature Jacobian. Then

$$s_{\min}(J_f(x)) \geq \nu^t \quad \text{and} \quad \|\nabla f(x)\| = \|J_f(x)^T w\| \geq \nu^t \|w\|$$

for every  $x \in \mathbb{R}^{d_0}$ .

(c) For all  $x, x' \in \mathbb{R}^{d_0}$ ,

$$\|J_f(x') - J_f(x)\|_{\text{op}} \leq S_2 M^2 \sqrt{t} \alpha^{2t-2} \|x' - x\|,$$

and therefore

$$\|\nabla f(x') - \nabla f(x)\| \leq \|w\| S_2 M^2 \sqrt{t} \alpha^{2t-2} \|x' - x\|.$$

In particular, if  $x_0$  also satisfies  $|f(x_0)| \leq a$  with

$$a \leq \frac{\nu^{2t} \|w\|}{6 S_2 M^2 \sqrt{t} \alpha^{2t-2}},$$

then conditions (A2) and (A3) of Theorem 4.1 hold at  $x_0$  with  $C_g = \nu^t \|w\|$ .

(d) Assume in addition that  $\sigma$  satisfies the pointwise nondegeneracy condition from [DM26],

$$\sigma'(r)^2 + \sigma'(-r)^2 \geq 2c_{\text{DM}}^2 \quad \text{for all } r \in \mathbb{R},$$

for some constant  $c_{\text{DM}} > 0$ . Then for every fixed input  $x \in \mathbb{R}^{d_0}$ , in expectation over the random initialization,

$$\mathbb{E}[\|J_f(x)\|_F^2] \geq \left(1 + \frac{\gamma_{\text{DM}}}{t}\right)^t \cdot d_0 \geq (1 + \gamma_{\text{DM}}) \cdot d_0,$$

where

$$\gamma_{\text{DM}} := 0.1(1 - e^{-0.005d_0})c_{\text{DM}}^2.$$

At Xavier initialization,  $M = O(1)$  with high probability by [Vershynin12], so for fixed depth  $t$  the constants in parts (b)–(c) are order one.

*Proof. Part (a).* For any unit vector  $v$ ,

$$\|A_\ell(x)v\| \geq \|v\| - \frac{1}{\sqrt{t}} \|D_\ell(x)\Lambda_\ell v\| \geq 1 - \frac{S_1 M}{\sqrt{t}} = \nu.$$

This gives  $s_{\min}(A_\ell(x)) \geq \nu$ . The upper bound is immediate from triangle inequality:

$$\|A_\ell(x)\|_{\text{op}} \leq 1 + \frac{1}{\sqrt{t}} \|D_\ell(x)\|_{\text{op}} \|\Lambda_\ell\|_{\text{op}} \leq 1 + \frac{S_1 M}{\sqrt{t}} = \alpha.$$

**Part (b).** Since  $J_f(x) = A_t(x) \cdots A_1(x)$ , submultiplicativity of the smallest singular value gives

$$s_{\min}(J_f(x)) \geq \prod_{\ell=1}^t s_{\min}(A_\ell(x)) \geq \nu^t.$$

Therefore

$$\|\nabla f(x)\| = \|J_f(x)^T w\| \geq s_{\min}(J_f(x)) \|w\| \geq \nu^t \|w\|.$$

**Part (c).** First, each residual block map

$$\Phi_\ell(h) := h + \frac{1}{\sqrt{t}} \sigma(\Lambda_\ell h)$$

is  $\alpha$ -Lipschitz by the operator-norm bound in part (a), so by induction

$$\|h_\ell(x') - h_\ell(x)\| \leq \alpha^\ell \|x' - x\| \quad (\ell = 0, \dots, t).$$

Next,

$$A_\ell(x') - A_\ell(x) = \frac{1}{\sqrt{t}} (D_\ell(x') - D_\ell(x)) \Lambda_\ell.$$

Because  $\sigma'$  is  $S_2$ -Lipschitz and  $\|\Lambda_\ell\|_{\text{op}} \leq M$ ,

$$\|D_\ell(x') - D_\ell(x)\|_{\text{op}} \leq S_2 \|\Lambda_\ell(h_{\ell-1}(x') - h_{\ell-1}(x))\| \leq S_2 M \alpha^{\ell-1} \|x' - x\|,$$

and hence

$$\|A_\ell(x') - A_\ell(x)\|_{\text{op}} \leq \frac{S_2 M^2}{\sqrt{t}} \alpha^{\ell-1} \|x' - x\|.$$

Now telescope the product:

$$J_f(x') - J_f(x) = \sum_{\ell=1}^t A_t(x') \cdots A_{\ell+1}(x') (A_\ell(x') - A_\ell(x)) A_{\ell-1}(x) \cdots A_1(x).$$

Taking operator norms and using  $\|A_j(\cdot)\|_{\text{op}} \leq \alpha$ ,

$$\|J_f(x') - J_f(x)\|_{\text{op}} \leq \sum_{\ell=1}^t \alpha^{t-\ell} \cdot \frac{S_2 M^2}{\sqrt{t}} \alpha^{\ell-1} \|x' - x\| \cdot \alpha^{\ell-1}.$$

Since  $\alpha^{t+\ell-2} \leq \alpha^{2t-2}$  for every  $\ell \leq t$ ,

$$\|J_f(x') - J_f(x)\|_{\text{op}} \leq S_2 M^2 \sqrt{t} \alpha^{2t-2} \|x' - x\|.$$

Multiplying by  $\|w\|$  gives the gradient bound. If  $|f(x_0)| \leq a$  and

$$a \leq \frac{\nu^{2t} \|w\|}{6 S_2 M^2 \sqrt{t} \alpha^{2t-2}},$$

then with  $C_g = \nu^t \|w\|$  and

$$H := \|w\| S_2 M^2 \sqrt{t} \alpha^{2t-2},$$

the  $C^2$  regularity of  $f$  and the global gradient-Lipschitz bound above imply  $\sup_z \|\nabla^2 f(z)\|_{\text{op}} \leq H$ , and we have  $H \leq C_g^2/(6a)$ . Part (b) gives (A2), Lemma 4.3 gives (A3), and the final claim follows.

**Part (d).** Fix an input  $x$  and write  $J = J_f(x) = A_t(x) \cdots A_1(x)$  for the corresponding feature Jacobian. Use the Rademacher identity  $\|J\|_F^2 = \mathbb{E}_r[\|Jr\|^2]$  for Rademacher  $r \in \{\pm 1\}^{d_0}$ . Define  $v_0 = r$  and  $v_\ell = A_\ell v_{\ell-1}$  for  $\ell = 1, \dots, t$ . Then

$$\|v_\ell\|^2 = \|v_{\ell-1}\|^2 + \frac{2}{\sqrt{t}} \langle v_{\ell-1}, D_\ell \Lambda_\ell v_{\ell-1} \rangle + \frac{1}{t} \|D_\ell \Lambda_\ell v_{\ell-1}\|^2. \quad (4)$$

*Cross-term cancellation (uses oddness of  $\sigma$ ).* The vector  $v_{\ell-1}$  is determined by layers  $1, \dots, \ell-1$  and is independent of layer  $\ell$ 's weights  $\Lambda_\ell$ . Each row  $\alpha_i$  of  $\Lambda_\ell$  is drawn i.i.d. from  $\mathcal{N}(0, d_0^{-1}I)$ . The  $i$ -th term of  $D_\ell \Lambda_\ell v_{\ell-1}$  is  $\sigma'(\alpha_i^T h_{\ell-1}) \cdot \alpha_i^T v_{\ell-1}$ . Since  $\sigma$  is odd,  $\sigma'$  is even, so the map  $\alpha \mapsto \sigma'(\alpha^T h) \alpha^T v$  is odd in  $\alpha$ . By Gaussian symmetry,

$$\mathbb{E}_{\alpha_i} [\sigma'(\alpha_i^T h) \alpha_i^T v] = 0,$$

and therefore

$$\mathbb{E}_{\Lambda_\ell} [\langle v_{\ell-1}, D_\ell \Lambda_\ell v_{\ell-1} \rangle \mid v_{\ell-1}, h_{\ell-1}] = 0.$$

*Quadratic term.* By [DM26, Lemma 6.1], applied with  $q = d_0$  Gaussian rows of variance  $1/d_0$ , we have

$$\|D_\ell \Lambda_\ell v_{\ell-1}\|^2 \geq 0.1 c_{\text{DM}}^2 \|v_{\ell-1}\|^2$$

with probability at least  $1 - e^{-0.005d_0}$  over  $\Lambda_\ell$ , conditional on  $v_{\ell-1}$  and  $h_{\ell-1}$ . Since the quadratic term is always nonnegative, taking expectation over the good event and its complement inserts the success-probability factor and gives the honest bound

$$\mathbb{E}_{\Lambda_\ell} [\|D_\ell \Lambda_\ell v_{\ell-1}\|^2 \mid v_{\ell-1}, h_{\ell-1}] \geq \gamma_{\text{DM}} \|v_{\ell-1}\|^2.$$

Combining with (4),

$$\mathbb{E}_{\Lambda_\ell} [\|v_\ell\|^2 \mid v_{\ell-1}, h_{\ell-1}] \geq \left(1 + \frac{\gamma_{\text{DM}}}{t}\right) \|v_{\ell-1}\|^2.$$

Iterating over layers and then averaging over  $r$  yields

$$\mathbb{E} [\|J_f(x)\|_F^2] \geq \left(1 + \frac{\gamma_{\text{DM}}}{t}\right)^t d_0 \geq (1 + \gamma_{\text{DM}}) d_0,$$

where the last inequality is Bernoulli's inequality.  $\square$

**Corollary 5.7** (Constructive one-step attacks for fixed-depth ResNets). *Assume additionally that inputs are drawn from the uniform measure on  $\mathbb{S}^{d-1}(R)$ , that the realized initialization satisfies  $\nu > 0$ , and that  $\mu(f = 0) = 0$  on  $\mathbb{S}^{d-1}(R)$ . Let*

$$L_f := \|w\| \alpha^t, \quad a := L_f \varepsilon_{\text{adv}}(1/2, \delta, d_0, R).$$

If

$$\varepsilon_{\text{adv}}(1/2, \delta, d_0, R) \leq \frac{\nu^{2t}}{6S_2 M^2 \sqrt{t} \alpha^{3t-2}},$$

then for a  $1 - \delta$  fraction of inputs  $x_0 \in \mathbb{S}^{d-1}(R)$ , the single gradient step

$$x_1 = x_0 - \text{sign}(f(x_0)) \frac{2a}{\|\nabla f(x_0)\|^2} \nabla f(x_0)$$

satisfies  $\text{sign}(f(x_1)) \neq \text{sign}(f(x_0))$  and

$$\|x_1 - x_0\| \leq \frac{2a}{\nu^t \|w\|} = 2 \left(\frac{\alpha}{\nu}\right)^t \varepsilon_{\text{adv}}(1/2, \delta, d_0, R) = O\left(\frac{R}{\sqrt{d_0}}\right).$$

*Proof.* By Proposition 4.10,  $\text{sign}(f)$  is exactly  $1/2$ -balanced under the uniform sphere measure. By Proposition 5.1 and Proposition 5.6(a),  $f$  is  $L_f$ -Lipschitz. Corollary 2.2 therefore gives  $|f(x_0)| \leq a$  for a  $1 - \delta$  fraction of inputs. On this set, Proposition 5.6 gives conditions (A2) and (A3), so Theorem 4.1 applies. The step-size bound follows from Proposition 5.6(b).  $\square$

*Remark 5.8* (The role of the  $1/\sqrt{t}$  scaling). The scaling  $1/\sqrt{t}$  in (3) serves two purposes. First, it ensures  $s_{\min}(A_\ell) \geq 1 - O(1/\sqrt{t}) > 0$ , preventing gradient vanishing. Second, it makes the cross term in (4) small ( $O(1/\sqrt{t})$ ), so that the identity contribution dominates. Without this scaling, the spectral norm of  $D_\ell \Lambda_\ell$  can exceed 1 (for  $d_0$  not too large), and the singular value lower bound fails. In practice, this scaling is implicitly achieved by careful initialization schemes [HeZRS16].

*Remark 5.9* (Relation to dynamical-isometry methods). There is a substantial large-width literature on Jacobian propagation in residual networks. In particular, Tarnowski et al. [Tarnowski19] show that ResNets can achieve dynamical isometry in a joint depth-width limit, with singular values concentrated near 1 at initialization. That perspective suggests a plausible route toward extending Proposition 5.6 beyond fixed depth. However, those asymptotic spectral-distribution results do not by themselves yield the pointwise Hessian control and compatibility estimate needed for Theorem 4.1. Bridging that gap remains an open problem.

### 5.3 Vision Transformers

We consider a simplified Vision Transformer (ViT), in the sense of the attention architecture introduced by Vaswani et al. [Vaswani17] and adapted to vision by Dosovitskiy et al. [Dosovitskiy21], consisting of  $L$  layers, each containing multi-head self-attention followed by an MLP block, with residual connections. We make the following structural assumptions: (i) all linear layers are bias-free, (ii) no positional encoding is added, (iii) any normalization layers (e.g., LayerNorm) are applied without affine parameters (i.e., no learnable shift or scale), and (iv) the activation function  $\sigma$  (in the MLP) is odd. Standard ViTs violate assumptions (i)–(iii); the proposition applies to this simplified variant.

**Proposition 5.10** (Oddness and balance for simplified ViTs; [Shmalo26, Section 5.4]). *Let  $f : \mathbb{R}^{n \times d} \rightarrow \mathbb{R}$  be a simplified ViT satisfying assumptions (i)–(iv) above. Then  $f(-X) = -f(X)$  for all inputs  $X$ , and hence  $\text{sign}(f)$  is exactly  $\frac{1}{2}$ -balanced under any input distribution  $\mu$  that is symmetric about the origin (i.e.,  $X \stackrel{d}{=} -X$ ), provided  $\mu(f = 0) = 0$ .*

*Proof.* We verify that each component of the transformer preserves oddness.

**Attention mechanism.** For an input  $X \in \mathbb{R}^{n \times d}$ , the queries, keys, and values are  $Q = XW_Q$ ,  $K = XW_K$ ,  $V = XW_V$ . Under negation:

$$Q(-X) = -XW_Q = -Q(X), \quad K(-X) = -K(X), \quad V(-X) = -V(X).$$

The attention logits are  $Q(-X)K(-X)^T = (-Q)(-K)^T = QK^T$ , so the attention weights  $A = \text{softmax}(QK^T/\sqrt{d_k})$  are *unchanged* by negation of the input. The attention output is

$$\text{Attn}(-X) = A \cdot V(-X) = A \cdot (-V(X)) = -\text{Attn}(X).$$

If the multi-head outputs are concatenated and passed through a bias-free output projection, oddness is preserved because the projection is linear.

**Normalization.** LayerNorm without affine parameters is odd: under  $X \mapsto -X$ , the token-wise mean changes sign while the centered second moment is unchanged, so the normalized output also changes sign. Thus such normalization layers do not break oddness.

**Residual connection.** If  $h$  is an odd function and  $g$  is an odd function, then  $h(X) + g(X)$  is odd. The residual connection  $X + \text{Attn}(X)$  is odd since both terms are.



**MLP block.** A bias-free MLP with odd activation  $\sigma$  computes  $\sigma(XW_1)W_2$ . Since  $\sigma$  is odd,  $\sigma(-XW_1) = -\sigma(XW_1)$ , so the MLP output is odd. The residual connection again preserves oddness.

**Readout.** A bias-free linear readout  $w^T h$  preserves oddness.

By induction over layers,  $f(-X) = -f(X)$ . Under any distribution symmetric about the origin (i.e.,  $X \stackrel{d}{=} -X$ ), the law of  $f(X)$  is symmetric about 0. If  $\mu(f = 0) = 0$ , then  $\mu(f > 0) = \mu(f < 0) = 1/2$ .  $\square$

*Remark 5.11 (Nondegeneracy).* The condition  $\mu(f = 0) = 0$  is generically satisfied: for a random network  $f$  with continuous weight distributions, the set  $\{X : f(X) = 0\}$  is a measure-zero hypersurface almost surely (under mild regularity assumptions on the input distribution  $\mu$ ).

To obtain a constructive theorem, we introduce a temperature parameter  $\lambda \geq 0$  in the attention logits and a residual scaling  $\beta = L^{-1/2}$ . We identify  $\mathbb{R}^{n \times d}$  with  $\mathbb{R}^{nd}$  and use the Frobenius norm / Euclidean norm interchangeably.

**Theorem 5.12** (High-temperature constructive regime for simplified ViTs). *Fix a compact convex set  $K \subset \mathbb{R}^{n \times d}$ . For each layer  $\ell = 1, \dots, L$ , define the attention map*

$$\text{Att}_{\lambda, \ell}(X) := P_{\lambda, \ell}(X) XW_{V, \ell} W_{O, \ell}, \quad P_{\lambda, \ell}(X) := \text{softmax}\left(\frac{\lambda XW_{Q, \ell}(XW_{K, \ell})^T}{\sqrt{d_k}}\right),$$

where the softmax is applied row-wise. Consider the residual stack

$$X_\ell = X_{\ell-1} + \beta \text{Att}_{\lambda, \ell}(X_{\ell-1}), \quad \beta = \frac{1}{\sqrt{L}},$$

and the scalar readout  $f(X) = \langle w, X_L \rangle$ . Let  $U = \frac{1}{n} \mathbf{1}\mathbf{1}^T$  and define the linear operators

$$B_\ell(H) := UHW_{V, \ell} W_{O, \ell}.$$

Then there exist constants  $C_1, C_2, \lambda_0 > 0$ , depending only on  $K$  and the realized weights, such that for every  $0 \leq \lambda \leq \lambda_0$  and every layer  $\ell$ :

$$(i) \sup_{X \in K} \|D\text{Att}_{\lambda, \ell}(X) - B_\ell\|_{\text{op}} \leq C_1 \lambda;$$

$$(ii) \sup_{X \in K} \|D^2\text{Att}_{\lambda, \ell}(X)\|_{\text{op}} \leq C_2 \lambda.$$

Fix  $0 \leq \lambda \leq \lambda_0$ , define

$$\Phi_{\lambda, \ell}(X) := X + \beta \text{Att}_{\lambda, \ell}(X), \quad \beta = \frac{1}{\sqrt{L}},$$

and assume

$$\Phi_{\lambda, \ell}(K) \subseteq K \quad \text{for every } \ell = 1, \dots, L.$$

If

$$B_0 := \max_{\ell} \|B_\ell\|_{\text{op}}, \quad \alpha_{\text{vit}} := 1 + \beta(B_0 + C_1 \lambda), \quad \nu_{\text{vit}} := 1 - \beta(B_0 + C_1 \lambda) > 0,$$

then for every  $X \in K$ , if we set  $X_0(X) := X$  and

$$X_\ell(X) := \Phi_{\lambda, \ell}(X_{\ell-1}(X)), \quad \ell = 1, \dots, L,$$

the full feature Jacobian is

$$J_{\text{vit}}(X) = A_{\lambda, L}(X_{L-1}(X)) \cdots A_{\lambda, 1}(X_0(X)), \quad A_{\lambda, \ell}(U) := D\Phi_{\lambda, \ell}(U),$$

and

$$\|\nabla f(X)\| \leq \alpha_{\text{vit}}^L \|w\|, \quad \|\nabla f(X)\| \geq \nu_{\text{vit}}^L \|w\|,$$

$$\|\nabla^2 f(X)\|_{\text{op}} \leq \|w\| L\beta C_2 \lambda \alpha_{\text{vit}}^{2L-2}.$$

In particular, if  $|f(X_0)| \leq a$  and

$$a \leq \frac{\nu_{\text{vit}}^{2L} \|w\|}{6L\beta C_2 \lambda \alpha_{\text{vit}}^{2L-2}},$$

with  $f(X_0) \neq 0$ , and the Frobenius ball

$$\{X : \|X - X_0\|_F \leq 2a/(\nu_{\text{vit}}^L \|w\|)\}$$

is contained in  $K$ , then conditions (A2) and (A3) hold at  $X_0$ , and the single gradient step of Theorem 4.1 crosses the decision boundary.

*Proof.* At  $\lambda = 0$ , the attention weights are exactly uniform:

$$P_{0,\ell}(X) = U \quad \text{for every } X \in K.$$

Hence

$$\text{Att}_{0,\ell}(X) = UXW_{V,\ell}W_{O,\ell}$$

is linear, so  $D\text{Att}_{0,\ell}(X) = B_\ell$  and  $D^2\text{Att}_{0,\ell}(X) = 0$ . Because the row-wise softmax is smooth, the map  $(X, \lambda) \mapsto \text{Att}_{\lambda,\ell}(X)$  is smooth on the compact set  $K \times [0, 1]$ . Hence the mixed derivatives  $\partial_\lambda D\text{Att}_{\lambda,\ell}(X)$  and  $\partial_\lambda D^2\text{Att}_{\lambda,\ell}(X)$  are uniformly bounded there. Using the fundamental theorem of calculus in the parameter  $\lambda$ ,

$$D\text{Att}_{\lambda,\ell}(X) - B_\ell = \int_0^\lambda \partial_s D\text{Att}_{s,\ell}(X) ds, \quad D^2\text{Att}_{\lambda,\ell}(X) = \int_0^\lambda \partial_s D^2\text{Att}_{s,\ell}(X) ds,$$

so there exist constants  $C_1, C_2, \lambda_0 > 0$  such that (i) and (ii) hold uniformly on  $K$  for all  $0 \leq \lambda \leq \lambda_0$ .

Let

$$A_{\lambda,\ell}(X) := D\Phi_{\lambda,\ell}(X).$$

Then

$$A_{\lambda,\ell}(X) = I + \beta(B_\ell + E_{\lambda,\ell}(X)), \quad \|E_{\lambda,\ell}(X)\|_{\text{op}} \leq C_1 \lambda.$$

Therefore

$$\text{smin}(A_{\lambda,\ell}(X)) \geq 1 - \beta(\|B_\ell\|_{\text{op}} + C_1 \lambda) \geq \nu_{\text{vit}},$$

and similarly  $\|A_{\lambda,\ell}(X)\|_{\text{op}} \leq \alpha_{\text{vit}}$ . Because each  $\Phi_{\lambda,\ell}$  maps  $K$  into itself, every trajectory point  $X_{\ell-1}(X)$  stays in  $K$  when  $X \in K$ . Therefore the bounds

$$\|D\Phi_{\lambda,\ell}(U)\|_{\text{op}} \leq \alpha_{\text{vit}}, \quad \text{smin}(D\Phi_{\lambda,\ell}(U)) \geq \nu_{\text{vit}}, \quad \|D^2\Phi_{\lambda,\ell}(U)\|_{\text{op}} \leq \beta C_2 \lambda$$

hold at every intermediate state  $U = X_{\ell-1}(X)$ . Applying the chain-rule recursion from the proof of Proposition 5.1 along this trajectory gives

$$\|J_{\text{vit}}(X)\|_{\text{op}} \leq \alpha_{\text{vit}}^L, \quad \text{smin}(J_{\text{vit}}(X)) \geq \nu_{\text{vit}}^L,$$

and

$$\|D^2 X_L(X)\|_{\text{op}} \leq L\beta C_2 \lambda \alpha_{\text{vit}}^{2L-2}.$$

Thus

$$\|\nabla f(X)\| = \|J_{\text{vit}}(X)^T w\| \leq \alpha_{\text{vit}}^L \|w\|, \quad \|J_{\text{vit}}(X)^T w\| \geq \nu_{\text{vit}}^L \|w\|,$$

and

$$\|\nabla^2 f(X)\|_{\text{op}} \leq \|w\| L\beta C_2 \lambda \alpha_{\text{vit}}^{2L-2}.$$

If  $|f(X_0)| \leq a$  and

$$a \leq \frac{\nu_{\text{vit}}^{2L} \|w\|}{6L\beta C_2 \lambda \alpha_{\text{vit}}^{2L-2}},$$

with  $f(X_0) \neq 0$ , then with  $C_g = \nu_{\text{vit}}^L \|w\|$  and

$$H = \|w\| L\beta C_2 \lambda \alpha_{\text{vit}}^{2L-2},$$

the  $C^2$  regularity of  $f$  and the gradient-Lipschitz bound above imply  $\sup_{X \in K} \|\nabla^2 f(X)\|_{\text{op}} \leq H$ , and we have  $H \leq C_g^2/(6a)$ . If moreover the Frobenius ball of radius  $2a/C_g$  around  $X_0$  is contained in  $K$ , then (A2) follows from the gradient lower bound above, (A3) follows from Lemma 4.3, and Theorem 4.1 applies.  $\square$

**Corollary 5.13** (Constructive one-step attacks for high-temperature simplified ViTs). *Assume the input law is the uniform measure on the sphere  $\mathbb{S}^{nd-1}(R_0) \subset \mathbb{R}^{nd}$ , assume  $\mu(f=0)=0$  on this sphere, and assume this sphere lies at distance at least*

$$\rho_0 := 2\left(\frac{\alpha_{\text{vit}}}{\nu_{\text{vit}}}\right)^L \varepsilon_{\text{adv}}(1/2, \delta, nd, R_0)$$

from the complement of  $K$ . Assume moreover that the residual blocks from Theorem 5.12 satisfy

$$\Phi_{\lambda, \ell}(K) \subseteq K \quad \text{for every } \ell = 1, \dots, L.$$

Let

$$L_f := \|w\| \alpha_{\text{vit}}^L.$$

If

$$\varepsilon_{\text{adv}}(1/2, \delta, nd, R_0) \leq \frac{\nu_{\text{vit}}^{2L}}{6L\beta C_2 \lambda \alpha_{\text{vit}}^{3L-2}},$$

then for a  $1 - \delta$  fraction of inputs  $X_0$  on  $\mathbb{S}^{nd-1}(R_0)$ , the single gradient step

$$X_1 = X_0 - \text{sign}(f(X_0)) \frac{2L_f \varepsilon_{\text{adv}}(1/2, \delta, nd, R_0)}{\|\nabla f(X_0)\|^2} \nabla f(X_0)$$

satisfies  $\text{sign}(f(X_1)) \neq \text{sign}(f(X_0))$  and

$$\|X_1 - X_0\|_F \leq 2\left(\frac{\alpha_{\text{vit}}}{\nu_{\text{vit}}}\right)^L \varepsilon_{\text{adv}}(1/2, \delta, nd, R_0) = O\left(\frac{R_0}{\sqrt{nd}}\right).$$

*Proof.* By Proposition 5.10,  $\text{sign}(f)$  is exactly  $1/2$ -balanced under the uniform sphere measure. Since the  $\rho_0$ -neighborhood of the sphere is contained in  $K$ , Theorem 5.12 gives the gradient upper bound  $\|\nabla f\| \leq L_f$  on that neighborhood, so in particular  $f$  is  $L_f$ -Lipschitz on the sphere. Corollary 2.2 therefore gives

$$|f(X_0)| \leq L_f \varepsilon_{\text{adv}}(1/2, \delta, nd, R_0)$$

for a  $1 - \delta$  fraction of inputs. The theorem also gives the gradient lower bound  $m = \nu_{\text{vit}}^L \|w\|$  and Hessian bound

$$H = \|w\| L\beta C_2 \lambda \alpha_{\text{vit}}^{2L-2}.$$

The displayed compatibility inequality is exactly  $L_f \varepsilon_{\text{adv}} \leq m^2/(6H)$ , and the distance assumption guarantees that the required attack neighborhood stays inside  $K$ . Therefore Theorem 4.9 applies.  $\square$

**Theorem 5.14** (Standard-temperature compact-set regime for simplified ViTs). *Let*

$$K_R := \{X \in \mathbb{R}^{n \times d} : \|X\|_F \leq R\}.$$

*Let  $S : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}$  denote the row-wise softmax map. Since  $S$  is smooth and acts row-wise on probability simplices, there exist universal constants*

$$c_{\text{sm},1} := \sup_{Z \in \mathbb{R}^{n \times n}} \|DS(Z)\|_{\text{op}} < \infty, \quad c_{\text{sm},2} := \sup_{Z \in \mathbb{R}^{n \times n}} \|D^2S(Z)\|_{\text{op}} < \infty.$$

*For each layer  $\ell = 1, \dots, L$ , define the standard-temperature attention map*

$$\text{Att}_\ell(X) := P_\ell(X) X M_\ell, \quad P_\ell(X) := S\left(\frac{XW_{Q,\ell}(XW_{K,\ell})^T}{\sqrt{d_k}}\right), \quad M_\ell := W_{V,\ell}W_{O,\ell},$$

*and set*

$$q_\ell := \|W_{Q,\ell}\|_{\text{op}}, \quad k_\ell := \|W_{K,\ell}\|_{\text{op}}, \quad m_\ell := \|M_\ell\|_{\text{op}}.$$

*Define*

$$L_{P,\ell} := \frac{2c_{\text{sm},1}Rq_\ell k_\ell}{\sqrt{d_k}}, \quad H_{P,\ell} := \frac{4c_{\text{sm},2}R^2q_\ell^2 k_\ell^2}{d_k} + \frac{2c_{\text{sm},1}q_\ell k_\ell}{\sqrt{d_k}},$$

$$\Gamma_\ell := m_\ell(\sqrt{n} + RL_{P,\ell}), \quad \Xi_\ell := m_\ell(RH_{P,\ell} + 2L_{P,\ell}).$$

*Then for every  $X \in K_R$ ,*

$$\|D\text{Att}_\ell(X)\|_{\text{op}} \leq \Gamma_\ell, \quad \|D^2\text{Att}_\ell(X)\|_{\text{op}} \leq \Xi_\ell.$$

*Now consider the residual stack*

$$X_\ell = X_{\ell-1} + \beta \text{Att}_\ell(X_{\ell-1}), \quad \beta = \frac{1}{\sqrt{L}},$$

*and define the residual blocks*

$$\Phi_\ell(X) := X + \beta \text{Att}_\ell(X).$$

*Assume*

$$\Phi_\ell(K_R) \subseteq K_R \quad \text{for every } \ell = 1, \dots, L.$$

*For  $X_0(X) := X$  and  $X_\ell(X) := \Phi_\ell(X_{\ell-1}(X))$ , let the scalar readout be  $f(X) = \langle w, X_L(X) \rangle$ . Let*

$$\Gamma := \max_\ell \Gamma_\ell, \quad \Xi := \max_\ell \Xi_\ell, \quad \alpha_{\text{std}} := 1 + \beta\Gamma, \quad \nu_{\text{std}} := 1 - \beta\Gamma.$$

*If  $\nu_{\text{std}} > 0$ , then for every  $X \in K_R$ ,*

$$J_{\text{std}}(X) = A_L(X_{L-1}(X)) \cdots A_1(X_0(X)), \quad A_\ell(U) := D\Phi_\ell(U),$$

$$\|\nabla f(X)\| \leq \alpha_{\text{std}}^L \|w\|, \quad \|\nabla f(X)\| \geq \nu_{\text{std}}^L \|w\|,$$

$$\|\nabla^2 f(X)\|_{\text{op}} \leq \|w\| L\beta\Xi \alpha_{\text{std}}^{2L-2}.$$

*In particular, if  $|f(X_0)| \leq a$ , with  $f(X_0) \neq 0$ , and*

$$a \leq \frac{\nu_{\text{std}}^{2L} \|w\|}{6L\beta\Xi \alpha_{\text{std}}^{2L-2}},$$

*and if the Frobenius ball*

$$\{X : \|X - X_0\|_F \leq 2a/(\nu_{\text{std}}^L \|w\|)\}$$

*is contained in  $K_R$ , then conditions (A2) and (A3) hold at  $X_0$ , and the single gradient step of Theorem 4.1 crosses the decision boundary.*

*Proof.* Let

$$Z_\ell(X) := \frac{XW_{Q,\ell}(XW_{K,\ell})^T}{\sqrt{d_k}}, \quad P_\ell(X) = S(Z_\ell(X)).$$

For  $H \in \mathbb{R}^{n \times d}$ ,

$$DZ_\ell(X)[H] = \frac{HW_{Q,\ell}(XW_{K,\ell})^T + XW_{Q,\ell}(HW_{K,\ell})^T}{\sqrt{d_k}}.$$

Because  $\|X\|_F \leq R$  on  $K_R$ ,

$$\|DZ_\ell(X)[H]\|_F \leq \frac{2Rq_\ell k_\ell}{\sqrt{d_k}} \|H\|_F.$$

Also

$$D^2Z_\ell(X)[H_1, H_2] = \frac{H_1W_{Q,\ell}(H_2W_{K,\ell})^T + H_2W_{Q,\ell}(H_1W_{K,\ell})^T}{\sqrt{d_k}},$$

so

$$\|D^2Z_\ell(X)[H_1, H_2]\|_F \leq \frac{2q_\ell k_\ell}{\sqrt{d_k}} \|H_1\|_F \|H_2\|_F.$$

By the chain rule,

$$DP_\ell(X)[H] = DS(Z_\ell(X))[DZ_\ell(X)[H]],$$

hence

$$\|DP_\ell(X)\|_{\text{op}} \leq c_{\text{sm},1} \frac{2Rq_\ell k_\ell}{\sqrt{d_k}} = L_{P,\ell}.$$

Similarly,

$$\begin{aligned} D^2P_\ell(X)[H_1, H_2] &= D^2S(Z_\ell(X))[DZ_\ell(X)[H_1], DZ_\ell(X)[H_2]] \\ &\quad + DS(Z_\ell(X))[D^2Z_\ell(X)[H_1, H_2]], \end{aligned}$$

so

$$\|D^2P_\ell(X)\|_{\text{op}} \leq c_{\text{sm},2} \left( \frac{2Rq_\ell k_\ell}{\sqrt{d_k}} \right)^2 + c_{\text{sm},1} \frac{2q_\ell k_\ell}{\sqrt{d_k}} = H_{P,\ell}.$$

Next,

$$\text{Att}_\ell(X) = P_\ell(X) X M_\ell.$$

Since each row of  $P_\ell(X)$  lies in the simplex,  $\|P_\ell(X)\|_F \leq \sqrt{n}$ , hence

$$\|P_\ell(X)\|_{\text{op}} \leq \sqrt{n}.$$

Differentiating,

$$D\text{Att}_\ell(X)[H] = DP_\ell(X)[H] X M_\ell + P_\ell(X) H M_\ell.$$

Therefore

$$\|D\text{Att}_\ell(X)[H]\|_F \leq L_{P,\ell} R m_\ell \|H\|_F + \sqrt{n} m_\ell \|H\|_F,$$

which gives

$$\|D\text{Att}_\ell(X)\|_{\text{op}} \leq \Gamma_\ell.$$

Differentiating once more,

$$\begin{aligned} D^2\text{Att}_\ell(X)[H_1, H_2] &= D^2P_\ell(X)[H_1, H_2] X M_\ell \\ &\quad + DP_\ell(X)[H_1] H_2 M_\ell + DP_\ell(X)[H_2] H_1 M_\ell. \end{aligned}$$

Hence

$$\|D^2\text{Att}_\ell(X)\|_{\text{op}} \leq m_\ell (RH_{P,\ell} + 2L_{P,\ell}) = \Xi_\ell.$$

Now define

$$A_\ell(X) := D\Phi_\ell(X).$$

The bound above implies

$$\|A_\ell(X) - I\|_{\text{op}} \leq \beta\Gamma,$$

so

$$\|A_\ell(X)\|_{\text{op}} \leq 1 + \beta\Gamma = \alpha_{\text{std}}, \quad s_{\min}(A_\ell(X)) \geq 1 - \beta\Gamma = \nu_{\text{std}}.$$

Because each  $\Phi_\ell$  maps  $K_R$  into itself, every trajectory point  $X_{\ell-1}(X)$  lies in  $K_R$  when  $X \in K_R$ . Hence the bounds

$$\|D\Phi_\ell(U)\|_{\text{op}} \leq \alpha_{\text{std}}, \quad s_{\min}(D\Phi_\ell(U)) \geq \nu_{\text{std}}, \quad \|D^2\Phi_\ell(U)\|_{\text{op}} \leq \beta\Xi$$

hold at every intermediate state  $U = X_{\ell-1}(X)$ . Applying the chain-rule recursion from the proof of Proposition 5.1 along this trajectory gives

$$\|J_{\text{std}}(X)\|_{\text{op}} \leq \alpha_{\text{std}}^L, \quad s_{\min}(J_{\text{std}}(X)) \geq \nu_{\text{std}}^L,$$

and

$$\|D^2X_L(X)\|_{\text{op}} \leq L\beta\Xi\alpha_{\text{std}}^{2L-2}.$$

Therefore

$$\begin{aligned} \|\nabla f(X)\| &= \|J_{\text{std}}(X)^T w\| \leq \alpha_{\text{std}}^L \|w\|, & \|J_{\text{std}}(X)^T w\| &\geq \nu_{\text{std}}^L \|w\|, \\ \|\nabla^2 f(X)\|_{\text{op}} &\leq \|w\| L\beta\Xi\alpha_{\text{std}}^{2L-2}. \end{aligned}$$

If

$$a \leq \frac{\nu_{\text{std}}^{2L} \|w\|}{6L\beta\Xi\alpha_{\text{std}}^{2L-2}},$$

then with

$$C_g = \nu_{\text{std}}^L \|w\|, \quad H = \|w\| L\beta\Xi\alpha_{\text{std}}^{2L-2},$$

we have  $H \leq C_g^2/(6a)$ . If moreover the Frobenius ball of radius  $2a/C_g$  around  $X_0$  lies in  $K_R$ , then (A2) follows from the gradient lower bound above, (A3) follows from Lemma 4.3, and Theorem 4.1 applies.  $\square$

**Corollary 5.15** (Sphere-based constructive theorem for standard-temperature simplified ViTs). *Assume the input law is the uniform measure on the sphere  $\mathbb{S}^{nd-1}(R_0) \subset \mathbb{R}^{nd}$  and assume  $\mu(f=0) = 0$  on this sphere. Fix  $R > R_0$ , let  $K_R = \{X \in \mathbb{R}^{n \times d} : \|X\|_F \leq R\}$ , and define  $\Gamma, \Xi, \alpha_{\text{std}}, \nu_{\text{std}}$  as in Theorem 5.14. Assume that the residual blocks from Theorem 5.14 satisfy*

$$\Phi_\ell(K_R) \subseteq K_R \quad \text{for every } \ell = 1, \dots, L.$$

Let

$$L_f := \|w\| \alpha_{\text{std}}^L, \quad \rho_{\text{std}} := 2 \left( \frac{\alpha_{\text{std}}}{\nu_{\text{std}}} \right)^L \varepsilon_{\text{adv}}(1/2, \delta, nd, R_0).$$

If  $\nu_{\text{std}} > 0$ ,

$$\varepsilon_{\text{adv}}(1/2, \delta, nd, R_0) \leq \frac{\nu_{\text{std}}^{2L}}{6L\beta\Xi\alpha_{\text{std}}^{3L-2}},$$

and  $\rho_{\text{std}} \leq R - R_0$ , then for a  $1 - \delta$  fraction of inputs  $X_0$  on  $\mathbb{S}^{nd-1}(R_0)$ , the single gradient step

$$X_1 = X_0 - \text{sign}(f(X_0)) \frac{2L_f \varepsilon_{\text{adv}}(1/2, \delta, nd, R_0)}{\|\nabla f(X_0)\|^2} \nabla f(X_0)$$

satisfies  $\text{sign}(f(X_1)) \neq \text{sign}(f(X_0))$  and

$$\|X_1 - X_0\|_F \leq \rho_{\text{std}} = O\left(\frac{R_0}{\sqrt{nd}}\right).$$

*Proof.* By Proposition 5.10,  $\text{sign}(f)$  is exactly  $1/2$ -balanced under the uniform sphere measure. The assumption  $\rho_{\text{std}} \leq R - R_0$  implies that the  $\rho_{\text{std}}$ -neighborhood of the sphere is contained in  $K_R$ . Theorem 5.14 therefore gives the gradient upper bound  $\|\nabla f\| \leq L_f$  on that neighborhood, so in particular  $f$  is  $L_f$ -Lipschitz on the sphere. Corollary 2.2 then yields

$$|f(X_0)| \leq L_f \varepsilon_{\text{adv}}(1/2, \delta, nd, R_0)$$

for a  $1 - \delta$  fraction of inputs on  $\mathbb{S}^{nd-1}(R_0)$ . The same theorem gives the gradient lower bound  $m = \nu_{\text{std}}^L \|w\|$  and Hessian bound

$$H = \|w\| L \beta \Xi \alpha_{\text{std}}^{2L-2}.$$

The displayed compatibility inequality is exactly  $L_f \varepsilon_{\text{adv}} \leq m^2/(6H)$ , and the assumption  $\rho_{\text{std}} \leq R - R_0$  guarantees that the required attack ball around a sphere point remains inside  $K_R$ . Therefore Theorem 4.9 applies.  $\square$

**Corollary 5.16** (Adding MLP residual sublayers to the standard-temperature ViT theorem). *For each layer  $\ell = 1, \dots, L$ , let*

$$Y_\ell = X_{\ell-1} + \beta \text{Att}_\ell(X_{\ell-1}), \quad X_\ell = Y_\ell + \beta G_\ell(Y_\ell),$$

where  $\text{Att}_\ell$  is the standard-temperature attention map from Theorem 5.14, and each  $G_\ell : \mathbb{R}^{n \times d} \rightarrow \mathbb{R}^{n \times d}$  is  $C^2$ . Assume that on  $K_R$ ,

$$\|DG_\ell(X)\|_{\text{op}} \leq \Lambda_\ell, \quad \|D^2G_\ell(X)\|_{\text{op}} \leq \Upsilon_\ell \quad \text{for every } X \in K_R.$$

Assume moreover that both residual sublayers preserve  $K_R$ :

$$X + \beta \text{Att}_\ell(X) \in K_R, \quad X + \beta G_\ell(X) \in K_R \quad \text{for every } X \in K_R \text{ and every } \ell.$$

Define

$$\begin{aligned} \alpha_\ell^{\text{att}} &:= 1 + \beta \Gamma_\ell, & \nu_\ell^{\text{att}} &:= 1 - \beta \Gamma_\ell, & H_\ell^{\text{att}} &:= \beta \Xi_\ell, \\ \alpha_\ell^{\text{mlp}} &:= 1 + \beta \Lambda_\ell, & \nu_\ell^{\text{mlp}} &:= 1 - \beta \Lambda_\ell, & H_\ell^{\text{mlp}} &:= \beta \Upsilon_\ell. \end{aligned}$$

If  $\nu_\ell^{\text{att}} > 0$  and  $\nu_\ell^{\text{mlp}} > 0$  for every  $\ell$ , then for the full simplified ViT readout  $f(X) = \langle w, X_L \rangle$  one has

$$\|\nabla f(X)\| \leq \|w\| \prod_{\ell=1}^L \alpha_\ell^{\text{att}} \alpha_\ell^{\text{mlp}}, \quad \|\nabla f(X)\| \geq \|w\| \prod_{\ell=1}^L \nu_\ell^{\text{att}} \nu_\ell^{\text{mlp}}$$

for every  $X \in K_R$ . Moreover,

$$\|\nabla^2 f(X)\|_{\text{op}} \leq \|w\| H_{\text{full}} \quad \text{for every } X \in K_R,$$

where

$$H_{\text{full}} := \sum_{r=1}^{2L} H_r \left( \prod_{j < r} \alpha_j \right)^2 \left( \prod_{j > r} \alpha_j \right),$$

and  $(\alpha_r, \nu_r, H_r)$  enumerate in order the  $2L$  residual sublayers

$$X \mapsto X + \beta \text{Att}_1(X), \quad X \mapsto X + \beta G_1(X), \quad \dots, \quad X \mapsto X + \beta \text{Att}_L(X), \quad X \mapsto X + \beta G_L(X).$$

Consequently, the same deterministic one-step conclusion as in Theorem 5.14 holds with the gradient lower bound and Hessian constant replaced by the effective full-block constants above. If, in addition, the full architecture satisfies the odd/bias-free assumptions of Proposition 5.10, then the same sphere-based conclusion follows exactly as in Corollary 5.15.

*Proof.* For the attention residual sublayer  $\Phi_\ell^{\text{att}}(X) := X + \beta \text{Att}_\ell(X)$ , Theorem 5.14 gives

$$\|D\Phi_\ell^{\text{att}}(X)\|_{\text{op}} \leq \alpha_\ell^{\text{att}}, \quad s_{\min}(D\Phi_\ell^{\text{att}}(X)) \geq \nu_\ell^{\text{att}}, \quad \|D^2\Phi_\ell^{\text{att}}(X)\|_{\text{op}} \leq H_\ell^{\text{att}}.$$

For the MLP residual sublayer  $\Phi_\ell^{\text{mlp}}(X) := X + \beta G_\ell(X)$ ,

$$D\Phi_\ell^{\text{mlp}}(X) = I + \beta DG_\ell(X), \quad D^2\Phi_\ell^{\text{mlp}}(X) = \beta D^2G_\ell(X),$$

so

$$\|D\Phi_\ell^{\text{mlp}}(X)\|_{\text{op}} \leq \alpha_\ell^{\text{mlp}}, \quad s_{\min}(D\Phi_\ell^{\text{mlp}}(X)) \geq \nu_\ell^{\text{mlp}}, \quad \|D^2\Phi_\ell^{\text{mlp}}(X)\|_{\text{op}} \leq H_\ell^{\text{mlp}}.$$

Because both residual sublayers preserve  $K_R$ , every intermediate feature state in the ordered attention/MLP trajectory remains in  $K_R$ . Applying the same chain-rule recursion as in Proposition 5.1 to that ordered list of  $2L$  residual sublayers proves the stated Jacobian and Hessian bounds. The constructive conclusion is then exactly Theorem 4.1 with the corresponding effective constants. If the full architecture is odd and bias-free, Proposition 5.10, Corollary 2.2, and Theorem 4.9 yield the same sphere-based conclusion as in Corollary 5.15.  $\square$

*Remark 5.17* (Standard-temperature ViT gradient conditions — open problem). Theorem 5.14 gives a deterministic standard-temperature compact-set theorem with explicit constants, so the remaining open problem is more specific than before. What is still missing is an automatic *random-initialization* verification of those constants on the relevant sphere neighborhood for standard Vision Transformers. The main difficulty is the attention mechanism: the Jacobian of the attention layer has both a “direct” contribution from the value matrix  $W_V$  and “indirect” contributions from the dependence of the attention weights  $A$  on the input through  $Q$  and  $K$ . Under standard transformer scaling ( $QK^\top/\sqrt{d_k}$ ), the attention logits are order-one random variables at initialization, so the softmax weights do *not* concentrate around  $1/n$  sharply enough for the perturbative argument of Theorem 5.12. The compact-set constants in Theorem 5.14 are always finite for fixed realized weights, but proving that they satisfy the useful regime  $\beta\Gamma < 1$  and the compatibility inequality at Xavier initialization on an  $O(R/\sqrt{nd})$  neighborhood remains open. This is consistent with the current regularity theory for attention: Kim, Papamakarios, and Mnih [Kim21] show that standard dot-product self-attention is not globally Lipschitz on unbounded domains, while Castin, Ablin, and Peyré [Castin24] prove local Lipschitz bounds on compact sets.

Corollary 5.15 shows that, once those deterministic constants are available on a suitable ambient ball, the usual sphere-based one-step conclusion follows exactly as in the high-temperature regime. We conjecture that conditions (A2)/(A3) hold for simplified ViTs at random initialization, and our numerical experiments (Section 9) support this. A plausible route is a two-scale argument on an  $O(R/\sqrt{nd})$  ambient neighborhood: first control the query/key logits and the softmax weights uniformly there by matrix concentration and empirical-process bounds, and then decompose the attention Jacobian into a direct value-path term plus a low-rank query/key correction coming from the softmax Jacobian. If one could show that this correction stays uniformly smaller than the identity-scale residual term with high probability, Theorem 5.14 would upgrade to a genuine random-initialization theorem.

*Remark 5.18* (Rigorous status across architectures). Let  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  be a random neural network at Xavier initialization, and let  $\tilde{f} : \mathbb{R}^d \rightarrow \mathbb{R}$  denote its natural extension to ambient space when needed (see Remark 4.8).

- (a) For convolutional networks satisfying the assumptions of [DM26] (smooth activation, constant depth, controlled width), conditions (A2) and (A3) are rigorous by Theorem 5.3, and condition (A1) is rigorous by [DM26, Lemma 8.1]. Hence the single gradient step is fully rigorous there.



- (b) Theorem 4.9 gives a model-independent route to constructive vulnerability once one verifies balance, a uniform gradient lower bound, and a curvature bound.
- (c) Proposition 5.1 gives one general way to verify those derivative bounds for smooth deep compositions.
- (d) For odd, bias-free feedforward/convolutional/residual networks under the uniform sphere measure, condition (A1) also follows from Proposition 4.10 and Theorem 2.1.
- (e) For the fixed-depth ResNet model (3), Proposition 5.6 gives rigorous deterministic bounds for (A2) and (A3); Corollary 5.7 therefore yields a full single-gradient-step theorem once the small-output radius is below the explicit compatibility threshold.
- (f) For simplified bias-free Vision Transformers, Proposition 5.10 gives exact balance under symmetric inputs when  $\mu(f = 0) = 0$ , Theorem 5.12 gives the local perturbative derivative bounds on a forward-invariant compact set, Corollary 5.13 turns them into a sphere-based constructive theorem in the high-temperature regime  $\lambda \ll 1$ , and Theorem 5.14 together with Corollaries 5.15 and 5.16 give deterministic standard-temperature compact-set / sphere theorems for fixed realized weights, including the usual attention-plus-MLP block structure.
- (g) For standard-temperature Vision Transformers, what remains open is the *automatic random-initialization verification* of the derivative constants required by Theorem 5.14 and Corollary 5.15 (Remark 5.17).

## 6 Multiclass adversarial fragility

We now extend the results from binary to multiclass classification. A  $K$ -class classifier is a measurable function  $F : \mathbb{S}^{d-1}(R) \rightarrow [K]$ . We say  $F$  is  $p$ -balanced if  $\mu(F(x) = j) \geq p$  for every class  $j \in \{1, \dots, K\}$ .

**Theorem 6.1** (Multiclass adversarial fragility). *Let  $F : \mathbb{S}^{d-1}(R) \rightarrow [K]$  be a measurable  $K$ -class classifier that is  $p$ -balanced with  $p > 0$ . Then for every  $\delta > 0$ ,*

$$\mu\{x : \exists y, \|y - x\| \leq \varepsilon_K(p, \delta), F(y) \neq F(x)\} \geq 1 - \delta,$$

where

$$\varepsilon_K(p, \delta) = r(p) + \frac{R}{\sqrt{d-1}} \sqrt{2 \log \left( \frac{KC_{\text{Lev}}}{\delta} \right)}. \quad (5)$$

*Proof.* Let  $A_k = \{x : F(x) = k\}$  for  $k = 1, \dots, K$ . By assumption,  $\mu(A_k) \geq p$  for every  $k$ . Define  $E_\varepsilon = \{x : \exists y, \|y - x\| \leq \varepsilon, F(y) \neq F(x)\}$ .

If  $x \in A_j \setminus E_\varepsilon$ , then the entire  $\varepsilon$ -ball around  $x$  is contained in  $A_j$ , so  $x \notin (A_k)_\varepsilon$  for every  $k \neq j$ . Therefore

$$\mathbb{1}_{E_\varepsilon^c}(x) \leq \sum_{k=1}^K \mathbb{1}_{(A_k)_\varepsilon^c}(x),$$

and integrating:  $\mu(E_\varepsilon^c) \leq \sum_{k=1}^K \mu((A_k)_\varepsilon^c)$ .

By Lévy's isoperimetric inequality on  $\mathbb{S}^{d-1}(R)$  [Ledoux01]: for any measurable  $A \subseteq \mathbb{S}^{d-1}(R)$  with  $\mu(A) \geq p$ ,

$$\mu(A_\varepsilon^c) \leq C_{\text{Lev}} \exp \left( - \frac{(d-1)(\varepsilon - r(p))_+^2}{2R^2} \right).$$

Setting  $\varepsilon = r(p) + \frac{R}{\sqrt{d-1}} \sqrt{2 \log(KC_{\text{Lev}}/\delta)}$  makes each term  $\leq \delta/K$ , so  $\mu(E_\varepsilon^c) \leq K \cdot (\delta/K) = \delta$ .  $\square$

**Remark 6.2** (No Lipschitz assumption needed). Unlike the margin-based approach (Corollary 6.3 below), Theorem 6.1 requires only measurability of  $F$ , not Lipschitz logits. This is the correct existential multiclass theorem: it guarantees a nearby class-changing perturbation, not merely a small margin.

**Corollary 6.3** (Margin bound for Lipschitz classifiers). *If additionally  $F = \arg \max_k f_k$  where each  $f_k : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  is  $L$ -Lipschitz, then the winning margin  $g_{j^*}(x) := f_{j^*}(x) - \max_{k \neq j^*} f_k(x)$  satisfies*

$$g_{j^*}(x) \leq 2L \varepsilon_K(p, \delta)$$

for a  $1 - \delta$  fraction of inputs.

*Proof.* If  $x$  has a class-changing neighbor  $y$  within distance  $\varepsilon$ , then by the  $2L$ -Lipschitz property of  $g_{j^*}$ :  $g_{j^*}(x) \leq g_{j^*}(x) - g_{j^*}(y) + g_{j^*}(y) \leq 2L\varepsilon + 0 = 2L\varepsilon$ , since  $g_{j^*}(y) \leq 0$  (the class changed at  $y$ ).  $\square$

*Remark 6.4* (Exchangeability gives balance only in expectation). Let  $F_\theta : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}^K$  be a random  $K$ -class network at initialization whose law is invariant under permutation of the output coordinates. Then for every class  $j$ ,

$$\mathbb{E}_\theta[\mu(\arg \max_k F_{\theta,k} = j)] = \frac{1}{K}.$$

This does not by itself imply that a fixed sampled network is  $1/K$ -balanced. Obtaining a high-probability fixed-network balance statement requires an additional concentration argument over the parameter randomness.

## 7 Beyond the sphere

The sphere  $\mathbb{S}^{d-1}(R)$  is a natural domain because the uniform measure on the sphere exhibits the strongest form of concentration. However, real-world data is typically not restricted to a sphere. We extend the main result to Gaussian inputs, which is arguably the most natural high-dimensional distribution.

**Theorem 7.1** (Gaussian inputs). *Let  $f : \mathbb{R}^d \rightarrow \mathbb{R}$  be 1-Lipschitz and let  $X \sim \mathcal{N}(0, I_d)$ . Suppose  $\text{sign}(f)$  is  $p$ -balanced under the Gaussian measure, i.e.,  $\mathbb{P}(f(X) > 0) \geq p$  and  $\mathbb{P}(f(X) < 0) \geq p$ . Then for every  $\delta \in (0, 1)$ :*

(i) *a  $1 - \delta$  fraction of inputs satisfy*

$$|f(X)| \leq r_G(p) + \sqrt{2 \log(2/\delta)},$$

*where  $r_G(p) = \sqrt{2 \log(1/p)}$ ;*

(ii) *a  $1 - \delta$  fraction of inputs admit  $Y$  with*

$$\|Y - X\| \leq r_G(p) + \sqrt{2 \log(2/\delta)}$$

*and  $\text{sign}(f(Y)) \neq \text{sign}(f(X))$ .*

*Proof.* Let  $A_+ = \{f > 0\}$  and  $A_- = \{f < 0\}$ , with Gaussian measures  $\gamma(A_+) \geq p$  and  $\gamma(A_-) \geq p$ . By the Gaussian isoperimetric inequality [BLM13], for any measurable  $A$  with  $\gamma(A) \geq p$ ,

$$\gamma(A_\epsilon) \geq \Phi(\Phi^{-1}(p) + \epsilon),$$

where  $\Phi$  is the standard normal CDF and  $A_\epsilon$  is the  $\epsilon$ -thickening. Applying this to  $A_+$  and  $A_-$ :

$$\gamma(A_{+, \epsilon}^c) \leq 1 - \Phi(\Phi^{-1}(p) + \epsilon), \quad \gamma(A_{-, \epsilon}^c) \leq 1 - \Phi(\Phi^{-1}(p) + \epsilon).$$

A point  $x$  lacks an adversarial neighbor within radius  $\epsilon$  only if  $x \notin A_{-, \epsilon}$  (when  $f(x) > 0$ ) or  $x \notin A_{+, \epsilon}$  (when  $f(x) < 0$ ). Hence  $\gamma(\text{no adversarial neighbor}) \leq \gamma(A_{+, \epsilon}^c) + \gamma(A_{-, \epsilon}^c) \leq 2(1 - \Phi(\Phi^{-1}(p) + \epsilon))$ . Setting each term to  $\delta/2$  and using  $\Phi^{-1}(1 - \delta/2) = |\Phi^{-1}(\delta/2)|$ , we need

$\Phi^{-1}(p) + \epsilon \geq |\Phi^{-1}(\delta/2)|$ , i.e.,  $\epsilon \geq |\Phi^{-1}(p)| + |\Phi^{-1}(\delta/2)|$ . Since  $|\Phi^{-1}(q)| \leq \sqrt{2\log(1/q)}$  for  $q < 1/2$ , the choice  $\epsilon = r_G(p) + \sqrt{2\log(2/\delta)}$  suffices. This proves part (ii).

For the output bound, let  $m_f$  be a median of  $f(X)$ , taking  $m_f = 0$  when  $p = 1/2$ . The Borell–Sudakov–Tsirelson inequality gives

$$\mathbb{P}(|f(X) - m_f| > t) \leq 2\exp(-t^2/2),$$

so taking  $t = \sqrt{2\log(2/\delta)}$  yields a  $1 - \delta$  event on which  $|f(X) - m_f| \leq t$ . It remains to bound the median. If  $m_f > 0$ , then the one-sided Gaussian concentration inequality around the median gives

$$p \leq \mathbb{P}(f(X) < 0) \leq \mathbb{P}(f(X) - m_f < -m_f) \leq \exp(-m_f^2/2),$$

hence  $m_f \leq \sqrt{2\log(1/p)} = r_G(p)$ . The case  $m_f < 0$  is symmetric, so  $|m_f| \leq r_G(p)$  in all cases. Combining the median bound with the deviation inequality proves part (i).  $\square$

*Remark 7.2* (Inputs on the ball). In [Shmalo26, Section 8] we discussed the extension to the ball  $B^d(R)$ . The concentration on the ball is weaker than on the sphere (the measure is spread over radius), but the adversarial radius remains  $O(R/\sqrt{d})$  with an additional logarithmic factor. We do not repeat the argument here.

## 8 Universality of adversarial vulnerability

A striking empirical phenomenon is the *transferability* of adversarial examples: an adversarial perturbation crafted for one model often fools a different model [GSS15, Papernot16]. The following theorem provides a partial theoretical explanation.

**Theorem 8.1** (Universality). *Let  $f, g : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  be two 1-Lipschitz classifiers with  $\text{sign}(f)$  being  $p_1$ -balanced and  $\text{sign}(g)$  being  $p_2$ -balanced. Then for every  $\delta > 0$ , a fraction at least  $1 - 2\delta$  of inputs  $x \in \mathbb{S}^{d-1}(R)$  are simultaneously adversarially vulnerable for both classifiers: there exist  $y_1, y_2$  with*

$$\begin{aligned} \|y_1 - x\| &\leq \varepsilon_{\text{adv}}(p_1, \delta, d, R), & \text{sign}(f(y_1)) &\neq \text{sign}(f(x)), \\ \|y_2 - x\| &\leq \varepsilon_{\text{adv}}(p_2, \delta, d, R), & \text{sign}(g(y_2)) &\neq \text{sign}(g(x)). \end{aligned}$$

Equivalently, after replacing  $\delta$  by  $\delta/2$ , one obtains a  $1 - \delta$  statement with radii  $\varepsilon_{\text{adv}}(p_i, \delta/2, d, R)$ .

*Proof.* Let  $f_1 = f$  and  $f_2 = g$ . Let  $E_i = \{x : \exists y, \|y - x\| \leq \varepsilon_{\text{adv}}(p_i, \delta, d, R), \text{sign}(f_i(y)) \neq \text{sign}(f_i(x))\}$  be the set of inputs with an adversarial neighbor for classifier  $f_i$ . By Theorem 2 of [Shmalo26],  $\mu(E_i) \geq 1 - \delta$  for  $i = 1, 2$ . Therefore  $\mu(E_1 \cap E_2) \geq 1 - \mu(E_1^c) - \mu(E_2^c) \geq 1 - 2\delta$ . The same argument extends to  $m$  classifiers:  $\mu(\bigcap_{i=1}^m E_i) \geq 1 - m\delta$ .  $\square$

**Corollary 8.2** (Simultaneous small output). *Under the same hypotheses, a  $1 - 2\delta$  fraction of inputs simultaneously satisfy  $|f(x)| \leq \varepsilon_{\text{adv}}(p_1, \delta, d, R)$  and  $|g(x)| \leq \varepsilon_{\text{adv}}(p_2, \delta, d, R)$ .*

*Proof.* Apply Theorem 2.1 to each classifier and take a union bound.  $\square$

**Theorem 8.3** (One-step transfer under directional alignment). *Let  $f, g : \mathbb{R}^d \rightarrow \mathbb{R}$  be  $C^2$ , and let  $x_0 \in \mathbb{R}^d$  satisfy  $f(x_0) > 0$  and  $g(x_0) > 0$ . Assume that  $x_0$  satisfies the hypotheses of Theorem 4.1 for  $f$  with parameter  $a_f$ , and define*

$$u_f := \frac{\nabla f(x_0)}{\|\nabla f(x_0)\|}, \quad \eta_f := \frac{2a_f}{\|\nabla f(x_0)\|}, \quad x_1 := x_0 - \eta_f u_f.$$

Assume moreover that for some  $m_{g|f}, H_g > 0$ ,

$$\langle \nabla g(x_0), u_f \rangle \geq m_{g|f}, \quad \sup_{\|z - x_0\| \leq \eta_f} \|\nabla^2 g(z)\|_{\text{op}} \leq H_g,$$

and that

$$g(x_0) < m_{g|f}\eta_f - \frac{H_g}{2}\eta_f^2.$$

Then  $f(x_1) < 0$  and  $g(x_1) < 0$ . In particular, the single perturbation  $x_1 - x_0$  is adversarial for both classifiers.

*Proof.* Theorem 4.1 gives  $f(x_1) < 0$ . For  $g$ , define

$$h(s) := g(x_0 - su_f), \quad s \in [0, \eta_f].$$

Then  $h'(0) = -\langle \nabla g(x_0), u_f \rangle \leq -m_{g|f}$ , and

$$h''(s) = u_f^T \nabla^2 g(x_0 - su_f) u_f$$

satisfies  $|h''(s)| \leq H_g$  on  $[0, \eta_f]$ . Taylor's theorem with remainder therefore gives

$$g(x_1) = h(\eta_f) \leq h(0) + \eta_f h'(0) + \frac{H_g}{2} \eta_f^2 \leq g(x_0) - m_{g|f} \eta_f + \frac{H_g}{2} \eta_f^2 < 0.$$

□

**Corollary 8.4** (Cosine-alignment transfer criterion). *Under the hypotheses of Theorem 8.3, suppose additionally that*

$$\|\nabla g(x_0)\| \geq C_g \quad \text{and} \quad \frac{\langle \nabla g(x_0), \nabla f(x_0) \rangle}{\|\nabla g(x_0)\| \|\nabla f(x_0)\|} \geq \kappa > 0.$$

If

$$g(x_0) < \kappa C_g \eta_f - \frac{H_g}{2} \eta_f^2,$$

then the same single gradient step for  $f$  also fools  $g$ .

*Proof.* The cosine-alignment bound implies

$$\langle \nabla g(x_0), u_f \rangle = \frac{\langle \nabla g(x_0), \nabla f(x_0) \rangle}{\|\nabla f(x_0)\|} \geq \kappa \|\nabla g(x_0)\| \geq \kappa C_g.$$

So Theorem 8.3 applies with  $m_{g|f} = \kappa C_g$ . □

*Remark 8.5* (Interpretation). Theorem 8.1 says that both classifiers are close to their respective decision boundaries at the same inputs. Theorem 8.3 strengthens this to a genuine transferability statement once one adds a directional transversality condition for the target classifier along the surrogate gradient direction. Thus the local quantities that govern transfer are not just small output and flatness, but also alignment of the target directional derivative with the source attack direction. This is consistent with recent transferability theory emphasizing that second-order effects alone are not the whole story [Fan24].

## 9 Numerical experiments

We present a single theorem-aligned CIFAR suite. The experiments focus on trained CIFAR-10 and CIFAR-100 models and evaluate exactly the local quantities that appear in Sections 4, 5.2, 5.3, and 8: margin size, margin-gradient norm, curvature compatibility, one-step success, and cross-model alignment. The three model families are a small ResNet, a plain CNN, and a compact ViT. Figure slots and full captions are prepared in Appendix B; the complete protocol is collected in Appendix A.

These numerics should be interpreted carefully. The theorems in the present paper concern balanced sphere-based models or explicitly controlled initialized architectures, whereas the CIFAR suite uses trained image classifiers on natural data. Accordingly, the experiments are not presented as proofs of the theorems in the trained setting. Their role is to show how the same local geometric quantities behave after training, and to separate phenomena that are directly tied to the constructive program of Remark 4.4 from generic attack-benchmarking statistics.

## 9.1 The four diagnostics

For a correctly classified test image  $x$  with label  $y$ , let  $c(x)$  be the runner-up class from the clean logits and define the smooth scalar margin

$$m(x) := \ell_y(x) - \ell_{c(x)}(x).$$

Keeping the competitor class fixed is important: it avoids the nonsmooth “max over classes” margin and makes the one-step Taylor expansion match Theorem 4.1. From this scalar margin we build four diagnostics.

- (N1) **Training trajectories.** During training we record test accuracy, a coarse predicted-class balance proxy  $\min_k \hat{\mu}(\arg \max_j \ell_j(x) = k)$ , one-step success for the theorem step  $x \mapsto x - 2m(x)\nabla m(x)/\|\nabla m(x)\|^2$ , and the fraction of sampled points whose empirical compatibility ratio is at most 1.
- (N2) **Radius calibration.** For each sampled image we compare the theorem-motivated predictor

$$r_{\text{pred}}(x) := \frac{2m(x)}{\|\nabla m(x)\|_2}$$

with an empirical radius obtained by binary search around an  $\ell_2$  PGD attack on the fixed runner-up margin. This compares the local first-order radius scale from Theorem 4.1 with the empirical attack radius on trained models.

- (N3) **Curvature compatibility.** Along the normalized margin-gradient direction  $u(x) = \nabla m(x)/\|\nabla m(x)\|$ , we estimate the directional second derivative by a centered finite difference and define

$$\rho_{\text{comp}}(x) := \frac{2m(x)\hat{H}(x)}{\|\nabla m(x)\|_2^2}.$$

The regime  $\rho_{\text{comp}}(x) \leq 1$  is the empirical analogue of the compatibility inequality in Remark 4.4. We also trace the margin along fractions of the theorem step to see how closely the local linear prediction matches the measured margin curve up to the decision crossing.

- (N4) **Transfer and alignment.** For every ordered pair of trained models, we apply the one-step perturbation generated by the source model and measure the target-model margin drop, transfer success, cosine between the two margin gradients, and the empirical slack in the sufficient inequality from Theorem 8.3. These are the numerical quantities corresponding to Theorem 8.3 and Corollary 8.4.

## 9.2 Questions addressed by the CIFAR suite

The CIFAR experiments address four descriptive questions. First, how early do one-step vulnerability signatures appear in training, and how persistent are they across architectures? Second, how conservative is the local radius scale  $2m/\|\nabla m\|$  relative to an actual  $\ell_2$  attack radius? Third, how does one-step success vary with the compatibility ratio and with the measured margin curve along the theorem step? Fourth, when perturbations transfer, how much structure is visible in gradient alignment and in the transfer-theorem slack?

Figure 1 gives the main theorem-aligned summary of the CIFAR suite. The dataset-specific training, radius, compatibility, and transfer figures, together with the full experimental protocol, are collected in Appendix B and Appendix A.

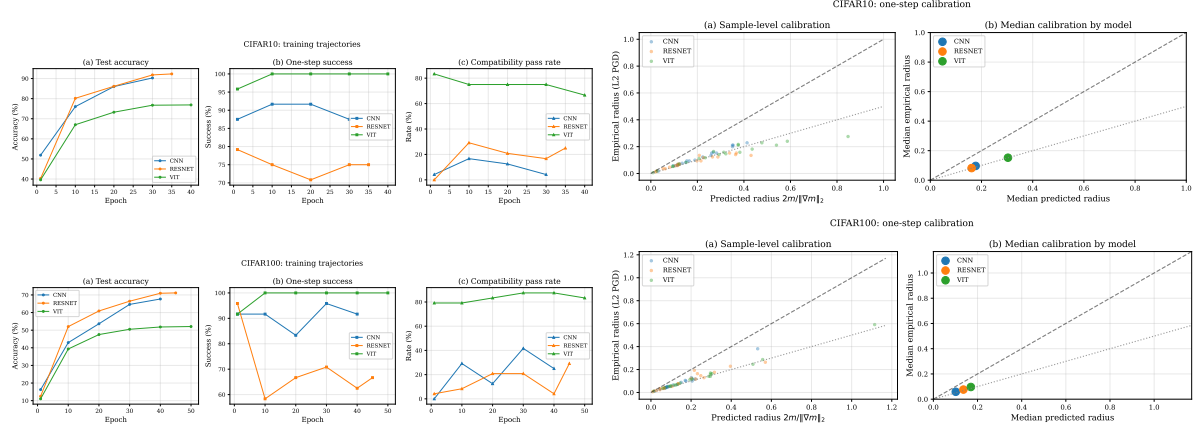


Figure 1: Main theorem-aligned CIFAR diagnostics. The top row reports the CIFAR-10 training trajectories and radius calibration, and the bottom row reports the corresponding CIFAR-100 figures. The left column compares ordinary test accuracy with the theorem-aligned one-step and compatibility diagnostics through training. The right column asks whether the first-order scale  $2m/\|\nabla m\|_2$  orders the true attack difficulty on trained models. The right-hand plots are the clearest empirical match to the theory: the empirical radius remains below the theorem step scale while still tracking it closely.

The easiest way to read the suite is by diagnostic strength. The training plots show that theorem-aligned local vulnerability signatures appear early and persist after the networks become accurate. The radius plots are the strongest quantitative match to the theory: they ask whether the theorem’s first-order step length is merely a proof artifact or whether it still reflects actual boundary distance after training. The compatibility plots are more qualitative: they probe whether the Taylor-expansion logic behind Remark 4.4 remains visible in trained models. The transfer plots are the most demanding and should be read as structure-revealing diagnostics rather than as exact calibrations of a sufficient condition.

Tables 1 and 2 report the final four-seed aggregates from the theorem-aligned sweep. Across both datasets, the ResNet has the highest accuracy, the CNN is slightly weaker but still competitive, and the compact ViT has the strongest one-step/compatibility signatures despite its lower baseline accuracy. Table 3 shows that the stronger ViT rerun substantially improves transformer accuracy without changing the qualitative theorem-aligned picture. In every configuration in Tables 1–3, the median predicted radius exceeds the median empirical radius, which is the expected conservative direction if the first-order theorem scale is informative but not exact after training. Across the saved runs, the radius comparison is the clearest match to the theory: on both CIFAR-10 and CIFAR-100, the empirical attack radius is strongly monotone in the predictor  $2m/\|\nabla m\|_2$  and stays below it, so the theorem scale behaves as a conservative first-order estimate rather than as an exact boundary distance. The compatibility figures are qualitatively consistent with the one-step Taylor picture, especially on CIFAR-100, but they are less clean than the radius plots. The transfer figures are the weakest numerically: alignment tracks margin drop more clearly than outright transfer success, and positive transfer-theorem slack appears only rarely in the dense sweeps. Accordingly, we treat the transfer numerics as qualitative structure checks rather than as a quantitative calibration of Theorem 8.3.

**Training trajectories.** The trajectory figures should be read by comparing the standard performance panel with the two local theorem-aligned panels. Accuracy improves steadily during training, but one-step success is already substantial very early and remains substantial at convergence. This matters because it shows that the constructive quantities are not simply proxies for poor accuracy. The ViT is the clearest example: in both datasets it is less accurate

than the ResNet while still exhibiting the strongest one-step signatures.

**Radius calibration.** The radius plots compare the theorem step length  $2m/\|\nabla m\|_2$  with a directly attacked radius. If the theorem scale had no geometric meaning after training, the scatter would be unstructured. Instead, the relation is strongly monotone on both datasets, and the empirical radii are consistently smaller. This is exactly the conservative direction one expects from a first-order sufficient step: the proof scale is informative, but it is not the exact boundary distance.

**Compatibility and proof ingredients.** The compatibility figures ask a subtler question. Small  $\rho_{\text{comp}}$  means that the curvature term is small relative to the first-order margin drop, so one expects the one-step argument to work better there. That trend is visible, especially on CIFAR-100, but the margin-curve panels also show that trained models are not literally linear along the whole path. The proof-ingredient boxplots explain those deviations by separating the roles of margin size, gradient norm, and directional curvature.

**Transfer and alignment.** The transfer panels should be interpreted cautiously. They do not show large absolute transfer rates. What they show is that the rare transfer events are not random: they cluster more strongly on model pairs with larger gradient cosine, and positive transfer-theorem slack acts as a sparse certification event rather than as a quantity that is frequently positive on natural trained models.

| model  | accuracy (%)     | one-step (%)      | compat. (%)      | pred. radius      | emp. radius       |
|--------|------------------|-------------------|------------------|-------------------|-------------------|
| ResNet | $92.17 \pm 0.17$ | $77.08 \pm 2.41$  | $20.83 \pm 5.89$ | $0.237 \pm 0.030$ | $0.110 \pm 0.006$ |
| CNN    | $90.41 \pm 0.20$ | $91.67 \pm 6.80$  | $9.38 \pm 7.12$  | $0.205 \pm 0.026$ | $0.114 \pm 0.014$ |
| ViT    | $77.03 \pm 0.42$ | $100.00 \pm 0.00$ | $63.54 \pm 7.89$ | $0.264 \pm 0.034$ | $0.136 \pm 0.013$ |

Table 1: CIFAR-10 theorem-aligned endpoint summary, aggregated over the four downloaded seeds from the baseline and extra sweeps. Accuracy is standard test accuracy. One-step and compatibility are the final theorem-step success rate and compatibility pass rate on correctly classified points. The last two columns report the per-seed median predicted radius  $2m/\|\nabla m\|_2$  and the corresponding empirical  $\ell_2$  attack radius. The main pattern is that the accuracy ordering and the theorem-aligned vulnerability ordering do not coincide across architectures.

| model  | accuracy (%)     | one-step (%)      | compat. (%)       | pred. radius      | emp. radius       |
|--------|------------------|-------------------|-------------------|-------------------|-------------------|
| ResNet | $71.58 \pm 0.50$ | $59.38 \pm 10.96$ | $25.00 \pm 18.00$ | $0.178 \pm 0.034$ | $0.102 \pm 0.022$ |
| CNN    | $67.95 \pm 0.30$ | $89.58 \pm 4.17$  | $22.92 \pm 7.22$  | $0.160 \pm 0.057$ | $0.096 \pm 0.035$ |
| ViT    | $52.20 \pm 0.32$ | $100.00 \pm 0.00$ | $81.25 \pm 4.17$  | $0.224 \pm 0.017$ | $0.120 \pm 0.008$ |

Table 2: CIFAR-100 theorem-aligned endpoint summary, aggregated over the four downloaded seeds from the baseline and extra sweeps. The higher class count lowers accuracy across all three architectures, but the same qualitative separation between accuracy and theorem-aligned one-step/compatibility behavior remains visible. The radius columns continue to show the conservative ordering predicted by the first-order picture.

| dataset   | run      | accuracy (%)     | one-step (%)      | compat. (%)      | pred. radius      | emp. radius       |
|-----------|----------|------------------|-------------------|------------------|-------------------|-------------------|
| CIFAR-10  | baseline | $77.03 \pm 0.42$ | $100.00 \pm 0.00$ | $63.54 \pm 7.89$ | $0.264 \pm 0.034$ | $0.136 \pm 0.013$ |
| CIFAR-10  | strong   | $85.59 \pm 0.38$ | $95.83 \pm 0.00$  | $8.33 \pm 5.89$  | $0.825 \pm 0.190$ | $0.211 \pm 0.029$ |
| CIFAR-100 | baseline | $52.20 \pm 0.32$ | $100.00 \pm 0.00$ | $81.25 \pm 4.17$ | $0.224 \pm 0.017$ | $0.120 \pm 0.008$ |
| CIFAR-100 | strong   | $65.06 \pm 0.23$ | $95.83 \pm 0.00$  | $70.83 \pm 5.89$ | $0.383 \pm 0.007$ | $0.186 \pm 0.002$ |

Table 3: Baseline versus stronger ViT runs. The stronger ViT setup substantially improves classification accuracy on both CIFAR-10 and CIFAR-100, but it does not remove the local theorem-aligned vulnerability signals measured by one-step success and by the gap between predicted and empirical radius. This table is included to show that the qualitative numerics are not an artifact of using an unusually weak transformer baseline.

## 10 Discussion

### 10.1 Summary of contributions

This paper bridges the geometric and dynamical approaches to adversarial vulnerability in deep neural networks. The main insight is that concentration of measure on the sphere (or under Gaussian inputs) provides a *general* bound on the output magnitude of balanced classifiers—the same bound that controls the adversarial radius. This replaces architecture-specific output calculations once balance has been established, for example via odd symmetry under a concentrated symmetric input law such as the uniform sphere measure.

The constructive gradient-step framework (Theorem 4.1) cleanly separates the geometric condition (small output, from balance) from the architectural conditions (gradient lower bound and stability, from the specific network structure). Corollary 2.2, Remark 4.2, Lemma 4.3, and Remark 4.4 turn this into a concrete proof program: small output, gradient lower bound, Hessian bound, compatibility. Theorem 4.9 packages this program into a model-independent statement in terms of balance, transversality, and curvature. Proposition 5.1 then gives a geometric mechanism for verifying those transversality/curvature conditions in deep compositional models. Corollary 5.2 packages this into a single constructive theorem for balanced smooth compositions. Proposition 4.10 records, following [Shmalo26], a large symmetry class where the geometric condition is automatic. For fixed-depth ResNets, Proposition 5.6 and Corollary 5.7 close the full one-step theorem. For simplified Vision Transformers, Proposition 5.10 proves balance, Theorem 5.12 controls the perturbative attention Jacobian on a forward-invariant compact set, Corollary 5.13 yields the sphere-based constructive theorem in the high-temperature regime, and Theorem 5.14 together with Corollaries 5.15 and 5.16 give deterministic standard-temperature compact-set / sphere theorems for fixed realized weights, including the usual attention-plus-MLP simplified ViT block. Finally, Theorem 8.3 and Corollary 8.4 turn the universality discussion into a concrete local transferability criterion.

### 10.2 Limitations

Our results apply to networks at *random initialization* (for the constructive gradient conditions) or to *any balanced Lipschitz classifier* (for the output bound). The random initialization setting is a strong assumption: trained networks have very different weight statistics. However, the balance condition often appears to persist at least qualitatively under training (as examined by the CIFAR experiments in Section 9.1, and as argued in [Shmalo26]), so the output bound of Theorem 2.1 remains a plausible organizing principle. The gradient conditions are less robust to training, and understanding when they hold for trained networks is an important open problem.

Our strongest new constructive statements are still regime-dependent. For ResNets, Proposition 5.6 closes the gap only in the fixed-depth residual-scaling regime; extending it to depths that grow with dimension would require sharper control of the product Jacobian and of the compatibility inequality in Remark 4.4. For Vision Transformers, Theorem 5.12 and Corollary 5.13



are perturbative in the attention temperature  $\lambda$  and require a compact neighborhood on which the local derivative bounds hold. Theorem 5.14 and Corollary 5.15 remove the temperature restriction for fixed realized weights, but the remaining hard step is still the random-initialization verification problem of Remark 5.17: the attention Jacobian’s dependence on the input through the softmax creates cross-terms that resist the layer-independence arguments available for feedforward and residual architectures.

### 10.3 Relation to other analytical methods

Several adjacent methods in the literature are stronger than parts of the present paper, but in different directions. For fully-connected random networks, Daniely and Shacham [DanielyS20], Bartlett et al. [BartlettBC21], and Bubeck et al. [BubeckCGT21] already give sharper architecture-specific theorems than we need here.

For residual networks, the closest stronger outside method is the dynamical-isometry / random-matrix approach, exemplified by Tarnowski et al. [Tarnowski19]. Those works control singular-value spectra in large-width limits and are the most plausible route toward a depth-growing extension of Proposition 5.6. What they do not currently provide is the local curvature control needed for our one-step theorem. For transformers, the current sharpest regularity tools remain the Lipschitz analyses of Kim et al. [Kim21] and Castin et al. [Castin24]; these justify our compact-set treatment and help explain Theorem 5.14, but they do not yet resolve the random-initialization verification problem for the standard-temperature constructive regime.

From the adversarial-transfer side, recent work such as Fan et al. [Fan24] studies how transferability depends on local geometry beyond naive flatness heuristics. Our Theorem 8.3 is compatible with that viewpoint: the right local quantity is a directional derivative of the target along the surrogate attack direction, together with a second-order error term.

### 10.4 Why the geometric–architectural separation is fundamental

The separation between the geometric condition (A1) and the architectural conditions (A2)/(A3) is not merely a technical convenience—it reflects a genuine mathematical barrier.

*Remark 10.1* (Why gradient lower bounds cannot follow from balance). One might hope to prove that *any* balanced Lipschitz function on the sphere has large gradient at most points, yielding a fully general constructive theorem. This is false. Consider  $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$  defined by  $f(x) = \epsilon \sigma(x_1/\epsilon)$  where  $\sigma$  is a smooth odd function with  $\sigma'(0) = 0$  and  $|\sigma'| \leq 1$ . This function is exactly balanced (by oddness) and  $O(1)$ -Lipschitz, but  $\|\nabla f(x)\| = |\sigma'(x_1/\epsilon)| = O(x_1/\epsilon)$ . For typical inputs on the high-dimensional sphere,  $|x_1| = O(R/\sqrt{d})$ , so  $\|\nabla f(x)\| = O(R/(\epsilon\sqrt{d}))$ , which can be made arbitrarily small by taking  $\epsilon$  large. The function is balanced, Lipschitz, and has small gradient at most points—yet adversarial examples still exist (by our Theorem 2.1 and [Shmalo26]). They simply cannot be found by a single gradient step.

*Remark 10.2* (Why Fourier diagonalization fails for nonlinear CNN features). A tempting approach to bounding the feature-correlation parameter  $\Gamma(a)$  for CNN feature maps is to use the Fourier block-diagonalization of [DM26]: each linear convolutional operator  $\Lambda_\ell$  decomposes as a direct sum over characters of the group  $G$ , with the blocks acting independently. However, pointwise nonlinearities (activation functions) *mix frequencies*: if  $a_\chi$  is the  $\chi$ -component of the input to a nonlinearity  $\sigma$ , the output  $\sigma(a)$  generally has nonzero projections onto all character components. This frequency mixing means the block-diagonal structure of the linear operators does not extend to the full nonlinear network, and a direct Fourier-based bound on  $\Gamma$  requires controlling the cross-frequency coupling through the nonlinearity—a genuinely difficult problem that we leave open.

## 10.5 Open questions

1. **Constructive attacks for arbitrary balanced classifiers.** Can one construct adversarial examples (not just prove existence) for *any* balanced Lipschitz classifier, without assuming differentiability or specific architecture? Our Theorem 4.1 requires gradient information; a black-box constructive result would be significantly stronger.
2.  **$\ell_\infty$  perturbations.** All our results use the  $\ell_2$  norm. The  $\ell_\infty$  setting, which is more common in practice, requires different concentration inequalities (e.g., for product measures rather than spherical measure). Can the same framework be extended?
3. **Non-abelian symmetry and equivariant networks.** Our oddness argument for ViTs (Proposition 5.10) uses the  $\mathbb{Z}/2$  symmetry  $x \mapsto -x$ . For equivariant networks (e.g., those respecting rotational or permutation symmetry), can the balance and vulnerability analysis be strengthened using the richer symmetry group?
4. **Trained networks.** The gradient non-degeneracy conditions (A2) and (A3) are now fully verified here for fixed-depth ResNets and, for simplified ViTs, under the forward-invariant compact-set hypotheses of Section 5.3; the trained-network setting remains largely open. Empirically, trained networks remain vulnerable. Can one prove gradient lower bounds for networks after training, perhaps using properties of the loss landscape?
5. **Tighter bounds for specific architectures.** Our general bound  $\varepsilon_{\text{adv}} = O(R/\sqrt{d})$  treats the network as a black-box Lipschitz function. Can architecture-specific analysis yield tighter bounds for particular network families?

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## Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used generative AI tools to accelerate drafting and revision. All mathematical claims, proofs, and bibliographic information were subsequently reviewed and edited by the author, who takes full responsibility for the content of the manuscript.

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## A Numerical protocol

All theorem-aligned CIFAR figures are generated by the script `cifar_rigorous_numerics.py`. The manuscript source, numerical code, and frozen artifacts are available at <https://github.com/yspennstate/adversarial-attacks>. The training runs for the paper use NVIDIA A40 GPUs. The same runner also supports dense post-processing from stored checkpoints via a `refresh-metrics` mode, so radius and transfer diagnostics can be recomputed at higher sample counts without retraining the model weights.

### A.1 Datasets and model families

We use the standard CIFAR-10 and CIFAR-100 train/test splits from `torchvision`. Training inputs use random crop with padding 4, random horizontal flip, and conversion to tensors. Test inputs are converted to tensors without augmentation. All models are wrapped with dataset-specific channelwise normalization:

$$\text{CIFAR-10 mean/std} = (0.4914, 0.4822, 0.4465)/(0.2470, 0.2435, 0.2616),$$

$$\text{CIFAR-100 mean/std} = (0.5071, 0.4867, 0.4408)/(0.2675, 0.2565, 0.2761).$$

The three model families used in the paper figures are:

- (a) a small ResNet with stage widths (32, 64, 128, 256) and two basic blocks per stage;
- (b) a plain CNN with channel widths (48, 96, 192);
- (c) a compact ViT with image size 32, patch size 4, 4 transformer layers, 4 heads, hidden dimension 192, and MLP dimension 384.

### A.2 Optimization and evaluation schedule

For the ResNet and CNN we use SGD with learning rate 0.1, momentum 0.9, Nesterov acceleration, weight decay  $5 \cdot 10^{-4}$ , and cosine annealing over the full epoch budget. For the ViT we use AdamW with learning rate  $3 \cdot 10^{-4}$ , weight decay 0.05, and cosine annealing.

The baseline figure sweeps use two seeds  $\{0, 1\}$ . The main endpoint tables in Section 9.1 aggregate those baseline runs together with matched extra sweeps at seeds  $\{2, 3\}$ , so the ResNet/CNN/ViT summaries there are four-seed aggregates. The training budgets are:

| dataset   | ResNet | CNN | ViT |
|-----------|--------|-----|-----|
| CIFAR-10  | 35     | 30  | 40  |
| CIFAR-100 | 45     | 40  | 50  |

The training runs use train batch size 256, test batch size 512, 8 dataloader workers, and evaluation every 10 epochs (plus epoch 1 and the final epoch). For each dataset/seed pair the dataloaders are rebuilt with seed-specific shuffling and worker seeding, so the reported seed variation reflects both model initialization and a reproducible data-order / augmentation stream rather than repeated reuse of a single loader state. The script stores both checkpoints and per-seed JSON metric summaries.

For a stronger ViT comparison, the same script also supports a `strong-vit` mode. That mode uses a larger compact ViT (6 layers, 8 heads, hidden dimension 256, MLP dimension 512, dropout/attention-dropout 0.1), RandAugment, label smoothing 0.1, AdamW with learning rate  $6 \cdot 10^{-4}$ , a short linear warmup followed by cosine decay, and longer budgets of 80 epochs on CIFAR-10 and 100 epochs on CIFAR-100. We report those runs separately because they sharpen the ViT comparison, not because they change the theorem-aligned diagnostics for the ResNet and CNN baselines.

### A.3 Local theorem-aligned diagnostics

All local diagnostics are computed only on correctly classified test images. For such an image  $x$  with true class  $y$ , let  $c(x)$  be the runner-up class under the clean logits and set

$$m(x) = \ell_y(x) - \ell_{c(x)}(x).$$

This fixed-competitor choice keeps the scalar margin smooth in a neighborhood of  $x$  and is therefore the correct object for matching Theorem 4.1.

The suite records the following quantities.

- (i) **One-step theorem attack.** The perturbation is

$$\Delta_{\text{thm}}(x) = \frac{2m(x)}{\|\nabla m(x)\|_2^2} \nabla m(x), \quad x_{\text{thm}} = \Pi_{[0,1]^d}(x - \Delta_{\text{thm}}(x)),$$

where  $\Pi_{[0,1]^d}$  denotes pixel clipping. Since  $m(x) > 0$  on correctly classified points, this is exactly the sign-corrected theorem step from Theorem 4.1 specialized to the fixed competitor margin.

- (ii) **Predicted radius.** We record

$$r_{\text{pred}}(x) = \frac{2m(x)}{\|\nabla m(x)\|_2},$$

which is the ambient  $\ell_2$  norm of the theorem step before clipping.

- (iii) **Empirical radius.** We estimate the smallest  $\ell_2$  radius at which the fixed-competitor margin can be driven below zero by binary search around an  $\ell_2$  PGD attack. In the reported experiments the binary search uses 9 radii, and each PGD solve uses 10 steps and 1 restart.
- (iv) **Compatibility ratio.** Let  $u(x) = \nabla m(x) / \|\nabla m(x)\|_2$ . Using a centered finite difference with step  $h = 10^{-3}$ , we estimate the directional second derivative

$$\hat{H}(x) := \frac{|m(x + hu(x)) - 2m(x) + m(x - hu(x))|}{h^2},$$

and define

$$\rho_{\text{comp}}(x) := \frac{2m(x)\hat{H}(x)}{\|\nabla m(x)\|_2^2}.$$

This is the empirical analogue of the compatibility condition in Remark 4.4. The raw values of  $m(x)$ ,  $\|\nabla m(x)\|_2$ ,  $\hat{H}(x)$ , and  $\rho_{\text{comp}}(x)$  are all stored, so the appendix can also include proof-ingredient plots that separate the contributions of (A1), (A2), and (A3) rather than only their final success/failure outcome.

- (v) **Margin curve.** For a small sample of points we evaluate the normalized margin along fractions  $\alpha \in [0, 1.25]$  of the theorem step:

$$\frac{m(x - \alpha \Delta_{\text{thm}}(x))}{|m(x)|}.$$

The comparison target is the linear prediction  $1 - 2\alpha$  from the first-order Taylor model. The predicted-class balance proxy recorded by the script is

$$\min_k \hat{\mu}(\arg \max_j \ell_j(x) = k)$$

over the full test set. This is only a coarse sanity check on class usage after training; it is *not* the theorem's exact balance parameter on the sphere.

## A.4 Transfer diagnostic

For each seed we form all ordered source/target pairs among the three trained architectures. We keep only test images classified correctly by every model in that seed’s trio. For a source model  $f$  and target model  $g$ , we apply the source theorem step  $\Delta_{\text{thm},f}(x)$  and record:

- (i) transfer success for  $g$ ;
- (ii) the target-margin drop  $m_g(x) - m_g(x - \Delta_{\text{thm},f}(x))$ ;
- (iii) the cosine

$$\cos(\nabla m_f(x), \nabla m_g(x)) = \frac{\langle \nabla m_f(x), \nabla m_g(x) \rangle}{\|\nabla m_f(x)\|_2 \|\nabla m_g(x)\|_2}.$$

- (iv) the directional derivative of the target margin along the source attack direction,

$$D_{u_f} m_g(x) = \left\langle \nabla m_g(x), \frac{\nabla m_f(x)}{\|\nabla m_f(x)\|_2} \right\rangle;$$

- (v) a centered finite-difference estimate  $\hat{H}_{g|f}(x)$  of the second derivative of  $m_g$  along that same source direction;
- (vi) the empirical transfer-theorem slack

$$s_{g|f}(x) := D_{u_f} m_g(x) \eta_f - \frac{\hat{H}_{g|f}(x)}{2} \eta_f^2 - m_g(x), \quad \eta_f := \frac{2m_f(x)}{\|\nabla m_f(x)\|_2}.$$

The cosine is the quantity from Corollary 8.4; the slack  $s_{g|f}(x)$  is the direct numerical counterpart of Theorem 8.3.

| source $\rightarrow$ target | C10 cosine | C10 transfer (%) | C100 cosine | C100 transfer (%) |
|-----------------------------|------------|------------------|-------------|-------------------|
| ResNet $\rightarrow$ CNN    | 0.140      | 8.98             | 0.110       | 6.25              |
| ResNet $\rightarrow$ ViT    | 0.024      | 3.13             | 0.018       | 2.08              |
| CNN $\rightarrow$ ResNet    | 0.140      | 3.91             | 0.110       | 3.13              |
| CNN $\rightarrow$ ViT       | 0.030      | 0.00             | 0.020       | 1.04              |
| ViT $\rightarrow$ ResNet    | 0.024      | 0.39             | 0.018       | 3.13              |
| ViT $\rightarrow$ CNN       | 0.030      | 0.00             | 0.020       | 2.08              |

Table 4: Pairwise transfer summary from the dense extra sweeps. Each entry averages the dense ordered source/target evaluations from the downloaded `cifar10_extra` and `cifar100_extra` outputs. The table should be read comparatively rather than absolutely: the transfer percentages are low overall, but the pairs with larger average cosine tend to have higher transfer rates, especially for the ResNet/CNN pair. This is the same qualitative ordering that appears in the transfer scatter plots.

## A.5 Figure-sweep budgets

The reported figure sweep uses

$$\text{trajectory-samples} = 12, \quad \text{radius-samples} = 10, \quad \text{transfer-samples} = 12.$$

These values were chosen to keep the full six-model training job tractable on a single A40 while still producing figure-ready diagnostics. The protocol itself does not depend on these particular values; rerunning the same script with larger diagnostic budgets only improves the sampling resolution. Additional high-resolution diagnostic passes simply reevaluate the same checkpoints with larger values such as `radius-samples=64` and `transfer-samples=128`. The proof-ingredient and transfer-theorem panels in Appendix B are taken from those higher-resolution diagnostic passes.

## B Additional figures

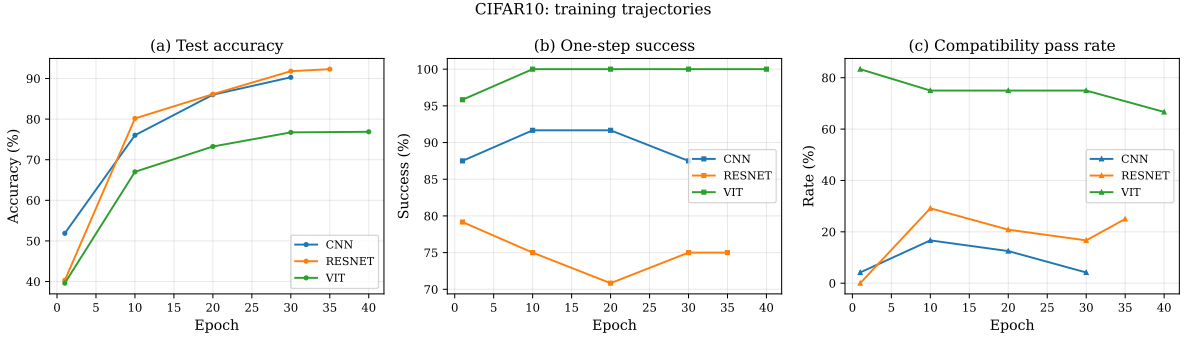


Figure 2: CIFAR-10 training trajectories for the ResNet, CNN, and ViT. Panel (a) is the standard test-accuracy curve. Panels (b) and (c) track the theorem-aligned local quantities: one-step success under the signed theorem step and the fraction of sampled points with  $\rho_{\text{comp}} \leq 1$ . The point of the figure is the comparison between these panels: one-step vulnerability is already substantial early in training and remains substantial even after the models become accurate, while compatibility remains architecture-dependent.

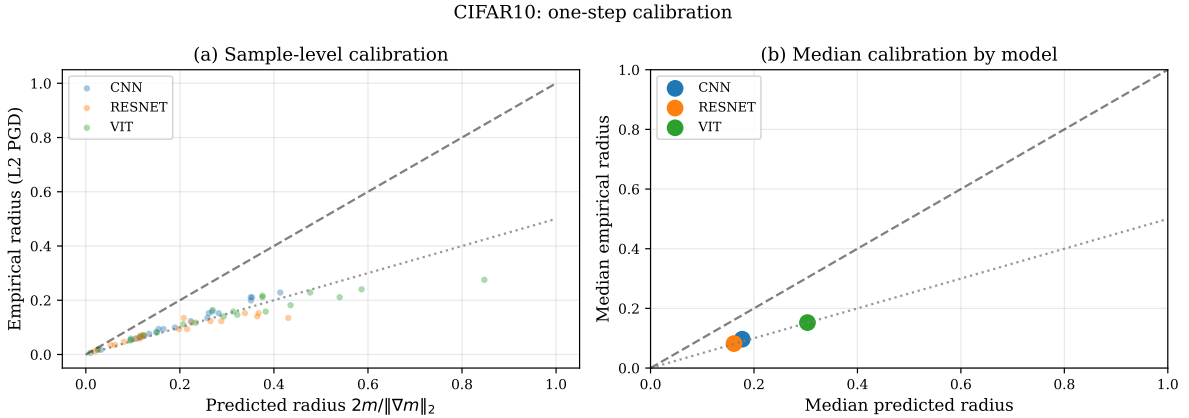


Figure 3: CIFAR-10 radius calibration. Panel (a) compares the theorem-predicted radius  $2m/\|\nabla m\|_2$  against the empirical  $\ell_2$  attack radius on individual correctly classified test images. Panel (b) repeats the comparison at the level of per-model medians. The dashed black line is  $y = x$ , and the dotted gray line is  $y = \frac{1}{2}x$ . Points below  $y = x$  mean that the theorem step length is larger than the measured boundary-crossing radius. The extra  $y = \frac{1}{2}x$  guide is included because, in a locally linear picture, the actual decision crossing is expected near  $m/\|\nabla m\|_2$ , i.e. at about half of the theorem step. The observed cloud follows exactly this conservative pattern: the predictor remains informative after training, but it is not an exact distance estimate.



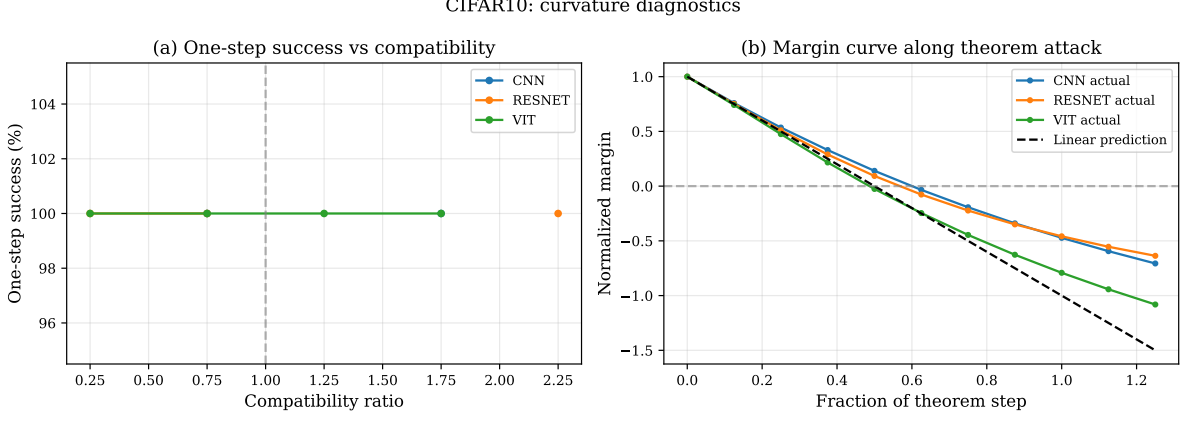


Figure 4: CIFAR-10 curvature diagnostics. Panel (a) bins one-step success by the empirical compatibility ratio  $\rho_{\text{comp}}$ ; smaller  $\rho_{\text{comp}}$  means that the quadratic remainder is smaller relative to the first-order margin drop. Panel (b) plots the normalized runner-up margin along fractions  $\alpha$  of the theorem step and compares it with the linear prediction  $1 - 2\alpha$  from the first-order Taylor model. The panels therefore separate two questions: whether smaller compatibility ratio is associated with easier one-step crossing, and how far the actual margin curve departs from the linearized picture before the decision boundary is reached.

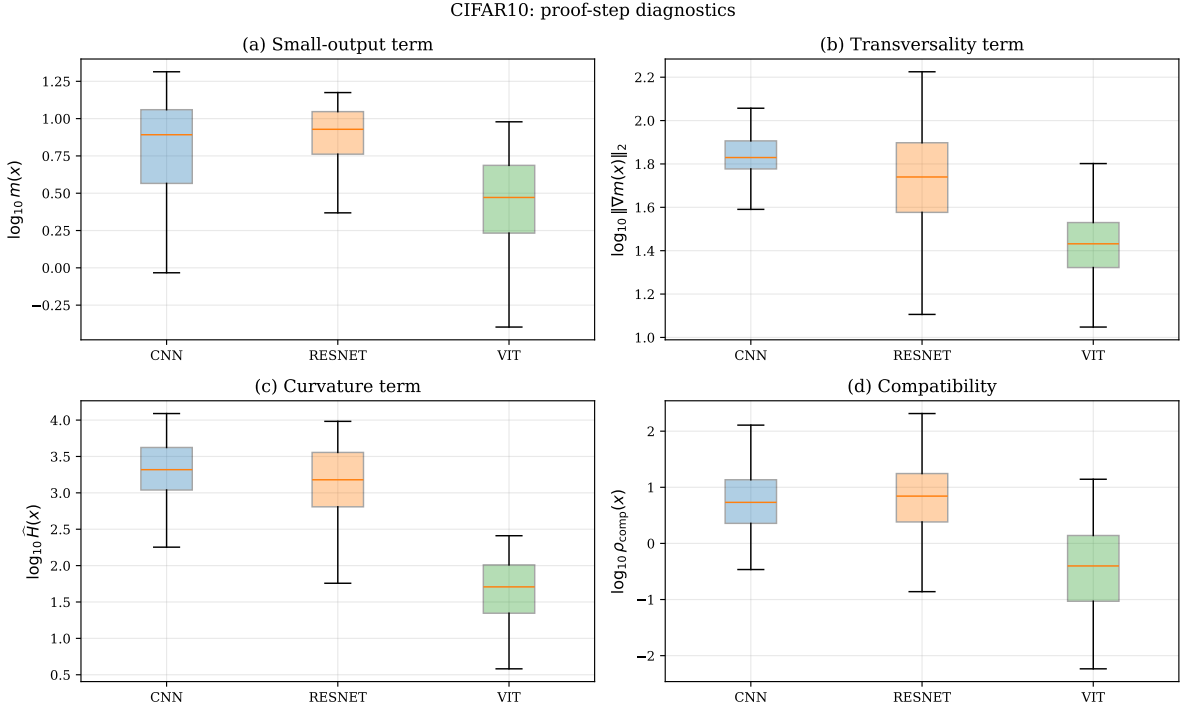


Figure 5: CIFAR-10 proof-step diagnostics. The panels show the empirical distributions of the runner-up margin  $m(x)$ , gradient norm  $\|\nabla m(x)\|_2$ , directional curvature estimate  $\hat{H}(x)$ , and compatibility ratio  $\rho_{\text{comp}}(x)$ . These are the four scalar ingredients that feed the constructive one-step picture. Larger margin or smaller gradient makes the predicted step longer, while larger directional curvature pushes the compatibility ratio upward. Reading the four boxplots together shows which part of the constructive mechanism is favorable or unfavorable for each architecture instead of compressing everything into a single success-rate number.

CIFAR10: transfer and alignment

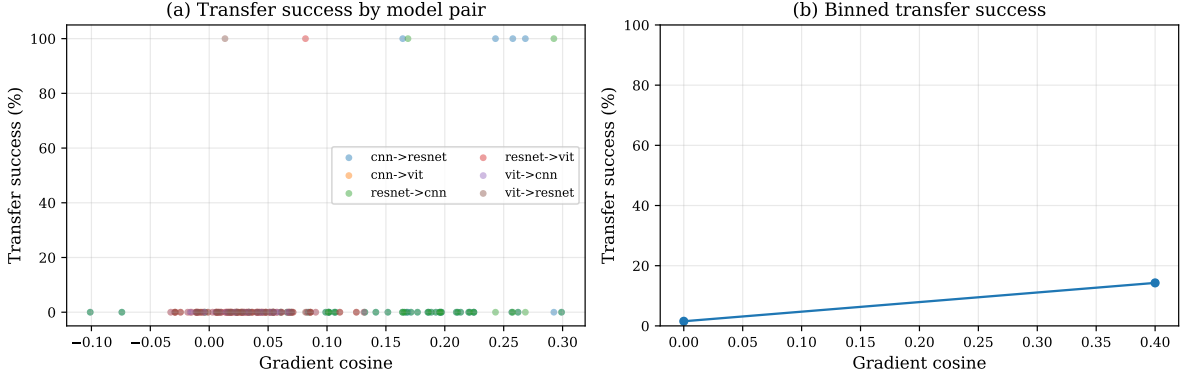


Figure 6: CIFAR-10 transfer and alignment. Panel (a) reports transfer success for every ordered source/target pair against the cosine between their margin gradients. Panel (b) shows the same relationship after cosine binning. Because the overall transfer rates are low, the figure should be read as a pair-ranking plot rather than as a dense regression plot: the few source/target pairs with visibly better transfer are also the pairs with larger gradient cosine.

CIFAR10: transfer theorem diagnostics

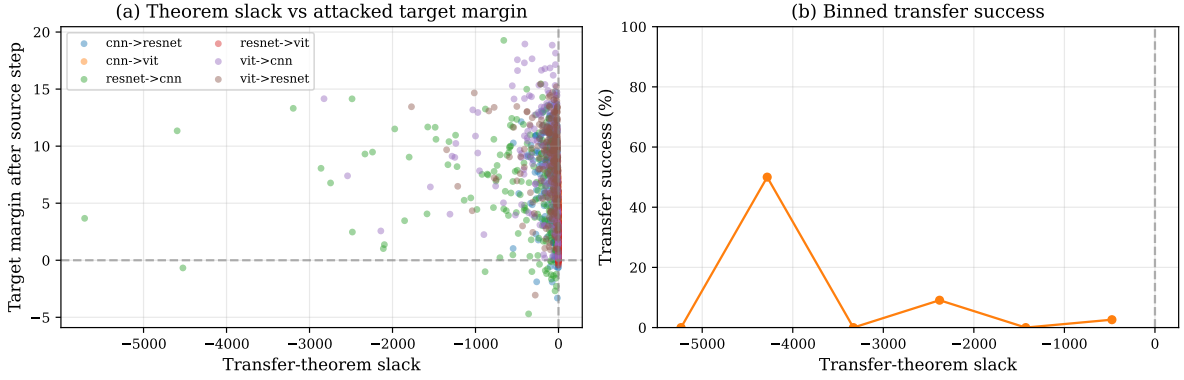


Figure 7: CIFAR-10 transfer-theorem diagnostics. Panel (a) compares the empirical slack from Theorem 8.3 with the target margin after the source-model step. Panel (b) bins transfer success by that slack. Positive slack is the regime in which the theorem’s sufficient inequality predicts a target-model crossing. In these sweeps that regime is rare, so the figure does not behave like a dense calibration plot. Instead it should be read as a sparse sufficiency diagnostic: the theorem is triggered only occasionally, but when it is triggered transfer becomes much more likely.

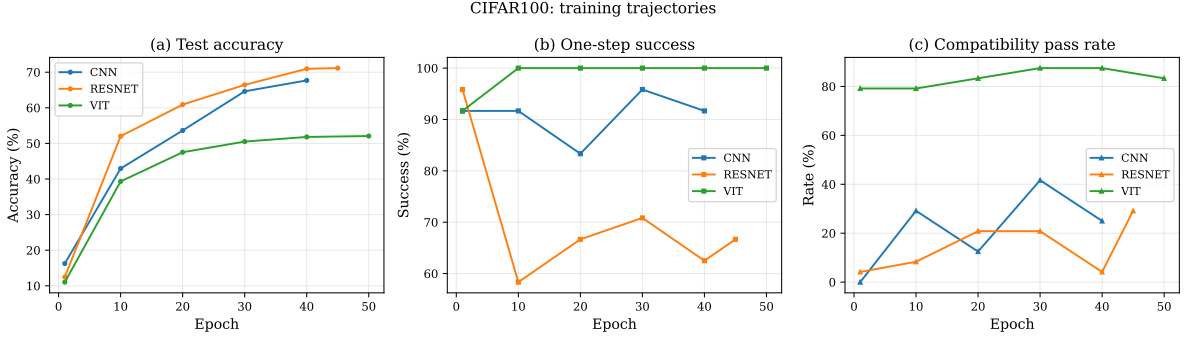


Figure 8: CIFAR-100 training trajectories for the ResNet, CNN, and ViT. As in Figure 2, panel (a) shows accuracy while panels (b) and (c) show the theorem-aligned one-step and compatibility diagnostics. The larger class count makes the margin problem harder, so this figure is the natural stress test for whether the same local quantities remain visible beyond CIFAR-10. The main qualitative pattern still survives: high accuracy does not eliminate strong one-step vulnerability signatures.

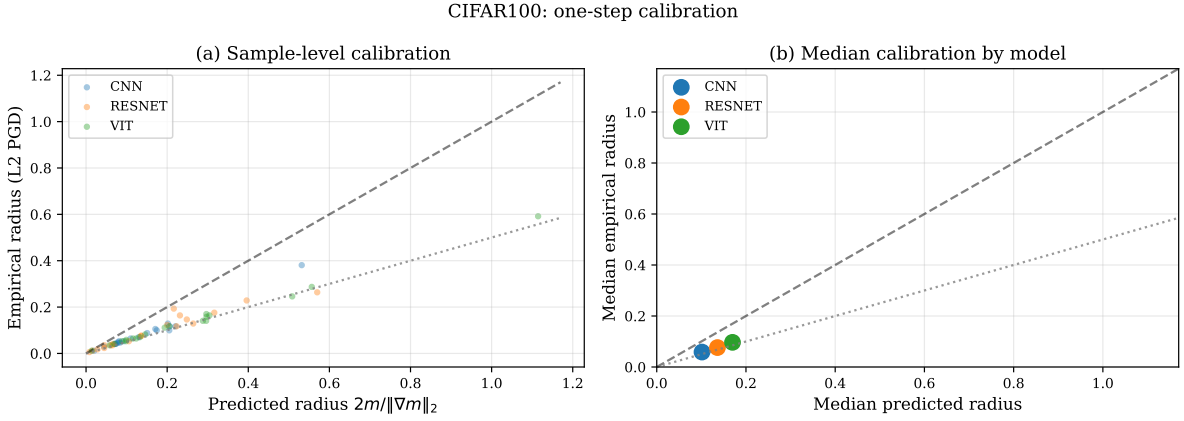


Figure 9: CIFAR-100 radius calibration. The sample-level and median-level comparisons are the CIFAR-100 analogue of Figure 3. As in CIFAR-10, the dashed black line is  $y = x$  and the dotted gray line is  $y = \frac{1}{2}x$ . The same interpretation applies here: points below  $y = x$  indicate that the theorem step is conservative, while proximity to  $y = \frac{1}{2}x$  is what one expects from a local linear boundary estimate. The striking point is robustness across datasets: even with 100 classes, the empirical radii remain strongly ordered by the theorem predictor.

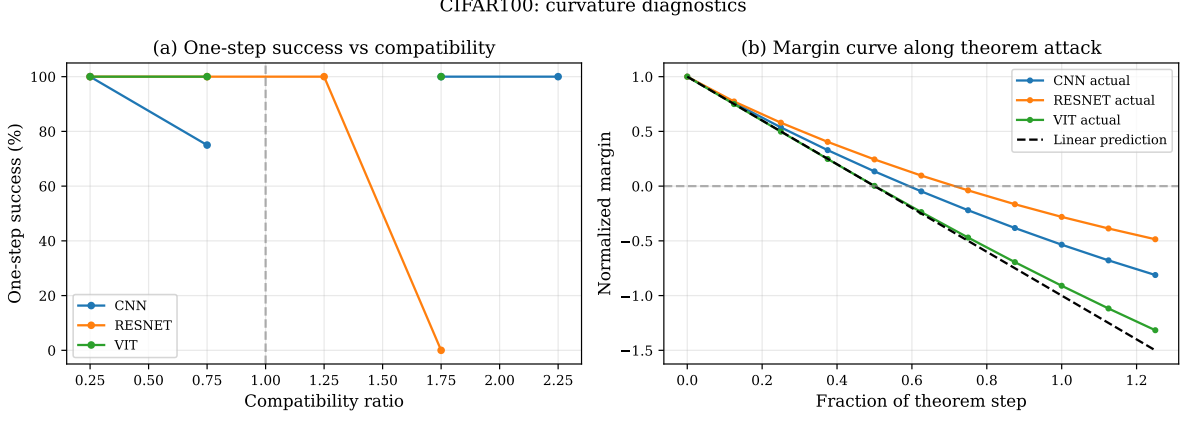


Figure 10: CIFAR-100 curvature diagnostics. The left panel bins one-step success by the empirical compatibility ratio, and the right panel compares the measured margin curve along the theorem step with the linear prediction. This is the CIFAR-100 counterpart of Figure 4, but it is often easier to read because the compatibility ratio separates successful and unsuccessful one-step behavior more clearly here. The right panel again shows that the theorem step usually reaches the crossing regime even though the path is not exactly linear.

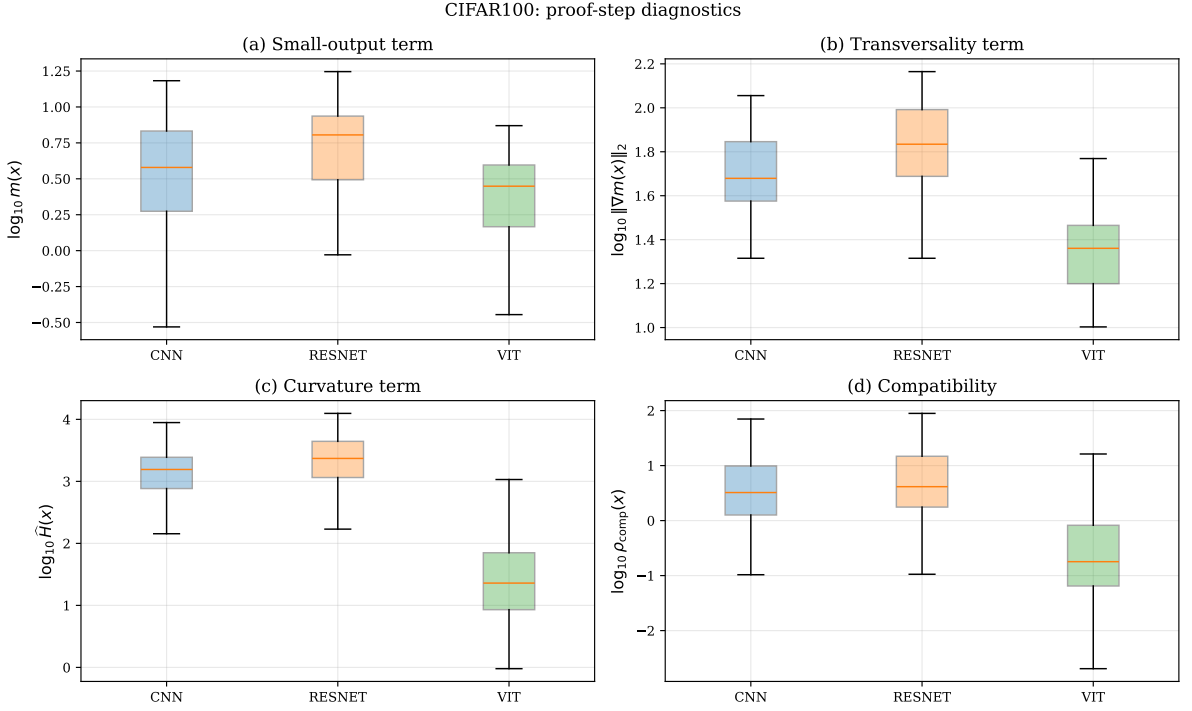


Figure 11: CIFAR-100 proof-step diagnostics. As in Figure 5, the panels separate the distributions of margin size, gradient norm, directional curvature, and compatibility ratio. The figure is descriptive rather than verdict-based. It shows how the same four proof ingredients are redistributed when the class count increases to 100, which helps explain why the compatibility and transfer panels look less clean in the harder dataset.

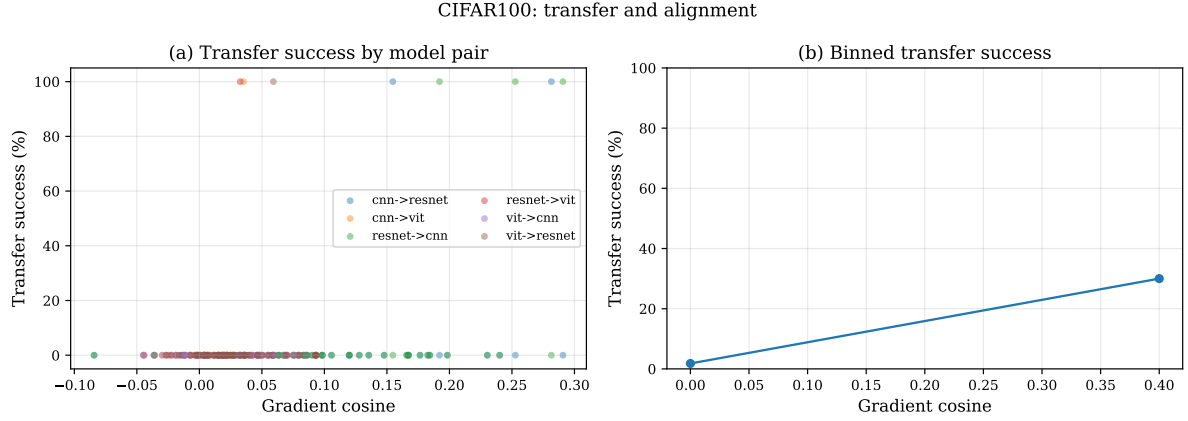


Figure 12: CIFAR-100 transfer and alignment. The source/target pair scatter and the binned cosine-success curve provide the CIFAR-100 counterpart of Figure 6. As in CIFAR-10, the overall transfer percentages are small, so the plot should be read comparatively. The point is not that cosine predicts transfer with high precision on every sample, but that the pairs with the best transfer signal are again the pairs with the larger average directional alignment.

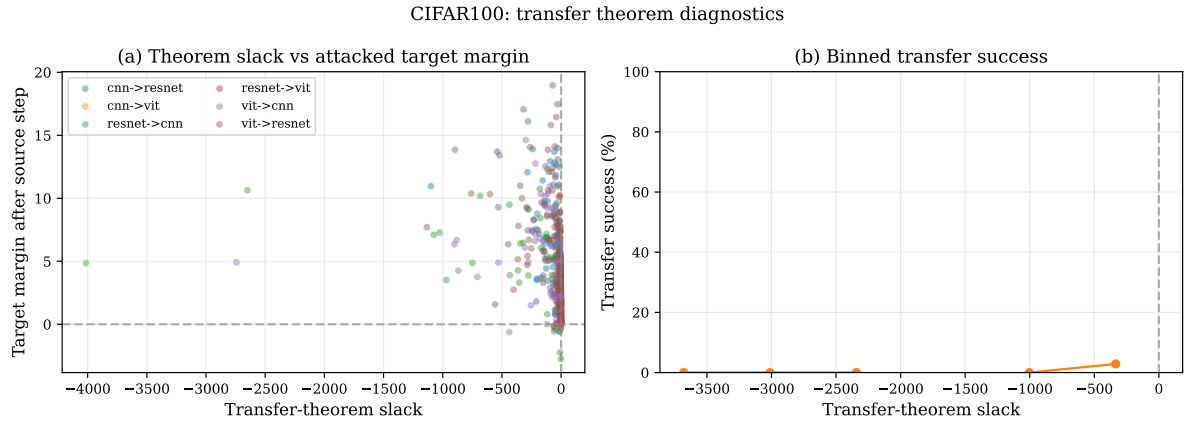


Figure 13: CIFAR-100 transfer-theorem diagnostics. As in Figure 7, the panels compare the empirical transfer-theorem slack with the attacked target margin and the transfer success rate. Positive slack is even rarer here than on CIFAR-10, so this figure again functions as a sparse sufficiency diagnostic rather than as a quantitative fit. The useful reading is binary: the theorem is almost never triggered in this harder setting, but when the slack does become positive it points in the correct direction.